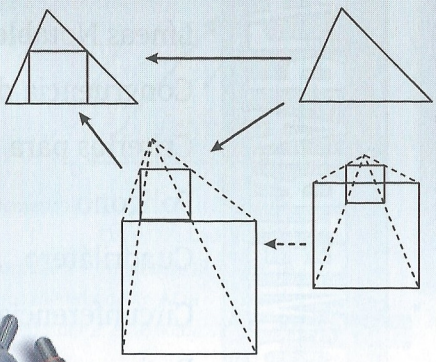
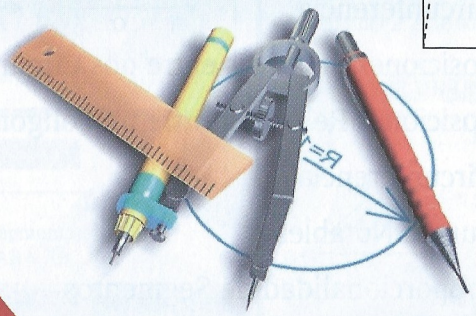
Resumen Teórico





Geometría Plana

FONDO EDITORIAL RODO

ÍNDICE

RESUMEN TEÓRICO DE GEOMETRÍA PLANA

Representación Geométrica: Segmentos – Ángulos
Triángulo
Líneas Notables asociadas al Triángulo
Congruencia de Triángulos
Criterios para hacer trazos auxiliares en Triángulos
Polígono
Cuadrilátero
Circunferencia
Posiciones Relativas entre dos Circunferencias
Posiciones Relativas entre un Polígono y una Circunferencia
Puntos Notables
Proporcionalidad de Segmentos
Semejanza de Triángulos
Relaciones Métricas en la Circunferencia
Relaciones Métricas en el Triángulo Rectángulo
Relaciones Métricas en Triángulos Oblicuángulos
Relaciones Métricas en Cuadriláteros
Polígono Regular
Áreas de Regiones Triangulares
Áreas de Regiones Cuadrangulares
Área del Círculo y sus Partes

REPRESENTACIÓN GEOMÉTRICA

Punto: A

Recta: $\overleftrightarrow{\mathscr{L}}$ (A, B)

Plano: P

Notación: Punto A

Notación: Recta $\mathscr{L} \equiv \overleftrightarrow{\mathscr{L}}$; Recta AB $\equiv \overleftrightarrow{AB}$

Notación: Plano P $\equiv \square$P

OBSERVACIÓN

* **Semirrecta**.- Porción en una recta determinada por un punto sin considerar a dicho punto.
* **Rayo**.- Es una semirrecta en donde se considera al punto de origen.

O: Origen del rayo.

Notación: Semirrecta OA $\equiv \overrightarrow{OA}$

Notación: Rayo OA $\equiv \overrightarrow{OA}$

SEGMENTO

A y B: Extremos del segmento.

Notación: Segmento AB $\equiv \overline{AB}$

MEDIDA DE UN SEGMENTO (LONGITUD)

b: Distancia entre A y B.

Notación:

Longitud de $\overline{AB} \equiv AB$

$\boxed{AB = b}$

PUNTO MEDIO DE UN SEGMENTO

Sea M el punto medio de AB, entonces:

$\boxed{AM = MB}$ $(M \in AB)$

Como las longitudes de los segmentos tienen valores enteros positivos, las operaciones entre ellas también origina valores positivos.

PUNTOS COLINEALES

Se denomina así cuando dos o más puntos pertenecen a una misma recta.

Ahora se observa:

$AB + BC = AC \rightarrow a + b = c$

$AB = AC - BC \rightarrow a = c - b$

ÁNGULO

Exterior — Interior

Elementos: O: Vértice del ángulo.
$\overrightarrow{OA}$ y $\overrightarrow{OB}$: Lados del ángulo.

Notación: Ángulo AOB $\equiv \angle AOB$

Si α: medida del $\angle$ AOB

Medida del $\angle$ AOB $\equiv m\angle AOB \rightarrow \boxed{m\angle AOB = \alpha}$

BISECTRIZ DE UN ÁNGULO

Si $\overrightarrow{OP}$: Ubicado en el interior del $\angle$AOB

Sea $\overrightarrow{OP}$ la bisectriz del $\angle$ AOB, por lo tanto se cumple que:

$\boxed{m\angle AOP = m\angle POB = \alpha}$

CLASIFICACIÓN DE LOS ÁNGULOS

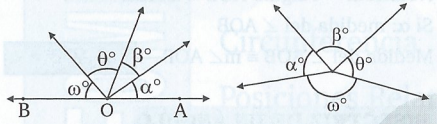
Ángulo Agudo	Ángulo Recto	Ángulo Obtuso
$0 < \alpha < 90$	$\alpha = 90$	$90 < \alpha < 180$

ÁNGULOS ADYACENTES Y ÁNGULOS CONSECUTIVOS

Un par de ángulos que compartan el mismo vértice y un lado se dice que son adyacentes.
Tres o más ángulos que compartan un lado dos a dos y un vértice se dice que son consecutivos. De la figura se observa.
$\angle AOB$ y $\angle BOC$ son ángulos adyacentes.
Los ángulos AOB, BOC, COD y DOE son ángulos consecutivos.
Además:

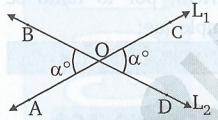
$$m\angle AOC = m\angle AOB + m\angle BOC$$
$$m\angle AOB = m\angle AOC - m\angle BOC$$
$$m\angle AOE = \alpha + \beta + \theta + \omega$$

CASOS ESPECIALES



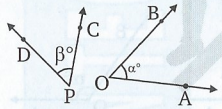
$\alpha + \beta + \theta + \omega = 180$ $\alpha + \beta + \theta + \omega = 360$

ÁNGULOS OPUESTOS POR EL VÉRTICE



$m\angle AOB = m\angle COD$
$m\angle BOC = m\angle AOD$

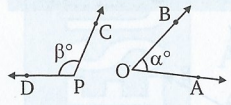
ÁNGULOS COMPLEMENTARIOS



$\alpha + \beta = 90$

Además:
α: Complemento de β;
$(C_\beta) \rightarrow C_\beta = 90 - \beta$
β: Complemento de α;
$(C_\alpha) \rightarrow C_\alpha = 90 - \alpha$

ÁNGULOS SUPLEMENTARIOS

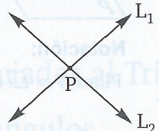


$\alpha + \beta = 180$

Además:
α: Suplemento de β;
$(S_\beta) \rightarrow S_\beta = 180 - \beta$
β: Suplemento de α;
$(S_\alpha) \rightarrow S_\alpha = 180 - \alpha$

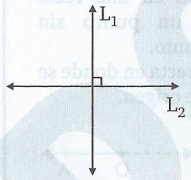
POSICIONES RELATIVAS ENTRE DOS RECTAS EN EL PLANO

RECTAS SECANTES



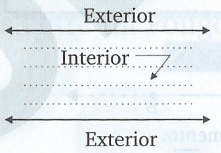
$\overleftrightarrow{L_1} \cap \overleftrightarrow{L_2} = \{P\}$
Entonces:
$\overleftrightarrow{L_1}$ y $\overleftrightarrow{L_2}$ son secantes
P: Punto de intersección

RECTAS PERPENDICULARES



En el gráfico:
$\overleftrightarrow{L_1} \perp \overleftrightarrow{L_2}$
⊥: "es perpendicular a"

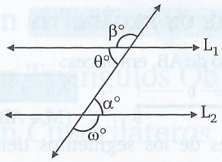
RECTAS PARALELAS



$\overleftrightarrow{L_1} \cap \overleftrightarrow{L_2} = \varnothing$
($\overleftrightarrow{L_1}$ y $\overleftrightarrow{L_2}$: coplanares)
Entonces:
$\overleftrightarrow{L_1} // \overleftrightarrow{L_2}$
//: "es paralela a"

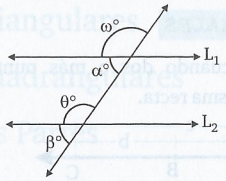
ÁNGULOS FORMADOS POR DOS RECTAS PARALELAS Y UNA SECANTE A ELLAS

A) ÁNGULOS ALTERNOS



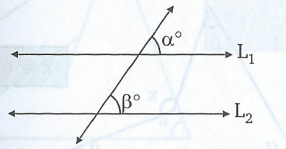
Si: $\overleftrightarrow{L_1} // \overleftrightarrow{L_2}$
Se cumple:
Ángulos Internos
$\alpha = \theta$
Ángulos Externos
$\beta = \omega$

B) ÁNGULOS CONJUGADOS



Si: $\overleftrightarrow{L_1} // \overleftrightarrow{L_2}$
Se cumple:
Ángulos Internos
$\alpha + \theta = 180°$
Ángulos Externos
$\beta + \omega = 180°$

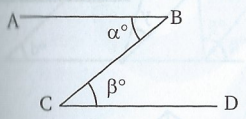
C) ÁNGULOS CORRESPONDIENTES



Si: $\overleftrightarrow{L_1} // \overleftrightarrow{L_2}$ Se cumple: $\alpha = \beta$

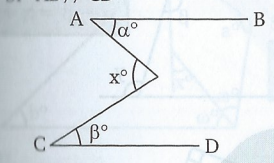
PROPIEDADES ADICIONALES

1. Si $\overline{AB} // \overline{CD}$



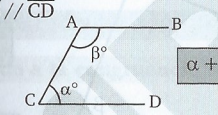
$\alpha = \beta$

2. Si $\overline{AB} // \overline{CD}$



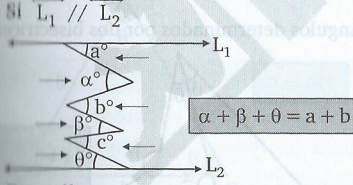
$x = \alpha + \beta$

3. Si $\overline{AB} // \overline{CD}$



$\alpha + \beta = 180$

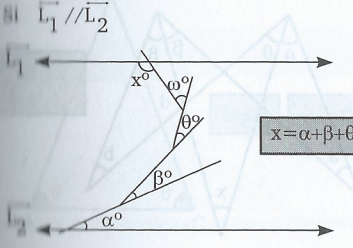
4. Si $\overleftrightarrow{L_1} // \overleftrightarrow{L_2}$



$\alpha + \beta + \theta = a + b + c$

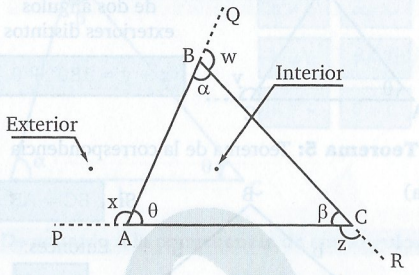
Esta última propiedad se cumple para cualquier número de lados.

5. Si $\overleftrightarrow{L_1} // \overleftrightarrow{L_2}$



$x = \alpha + \beta + \theta + \omega$

TRIÁNGULO

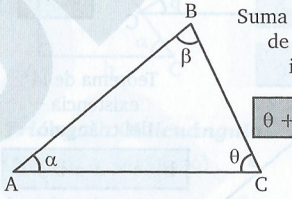


Elementos Vértices: A, B y C
Lados: $\overline{AB}$, $\overline{BC}$ y $\overline{AC}$

Notación ΔABC: Triángulo de vértices A, B y C

TEOREMAS FUNDAMENTALES

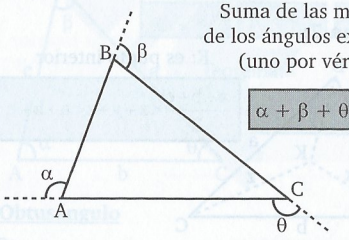
Teorema 1:



Suma de las medidas de los ángulos interiores

$\theta + \alpha + \beta = 180°$

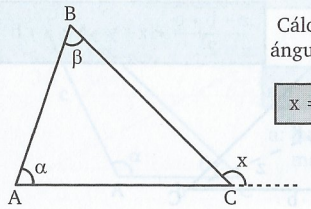
Teorema 2:



Suma de las medidas de los ángulos exteriores (uno por vértice)

$\alpha + \beta + \theta = 360°$

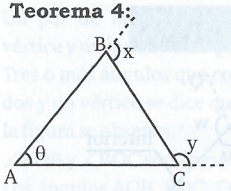
Teorema 3:



Cálculo de un ángulo exterior

$x = \alpha + \beta$

Teorema 4:

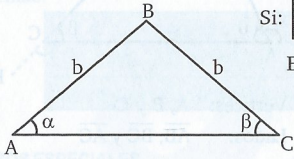


Suma de las medidas de dos ángulos exteriores distintos

$x + y = 180° + \theta$

Teorema 5: Teorema de la correspondencia

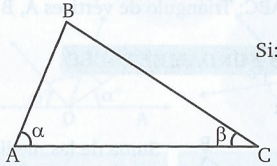
a)



Si: $BC = AB$

Entonces: $\alpha = \beta$

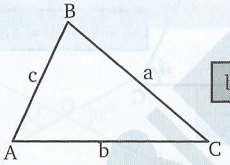
b)



Si: $BC > BA$

Entonces: $\alpha > \beta$

Teorema 6:

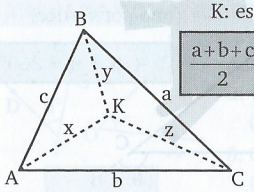


Teorema de la existencia del triángulo

$b - c < a < b + c$

$b \geq c$

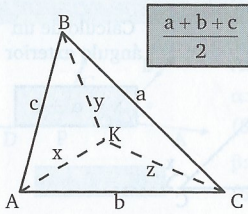
Teorema 7:



K: es punto interior

$\frac{a+b+c}{2} < x+y+z < a+b+c$

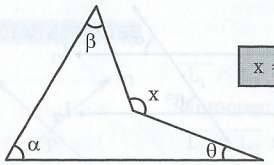
Teorema 8: Si: $a > b > c$



$\frac{a+b+c}{2} < x+y+z < a+b$

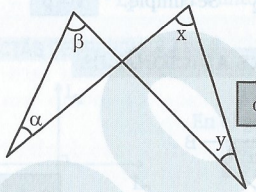
TEOREMAS ADICIONALES

1.



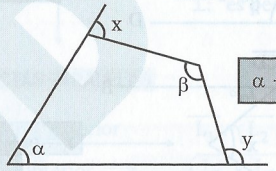
$x = \alpha + \beta + \theta$

2.



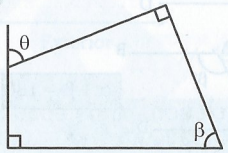
$\alpha + \beta = x + y$

3.



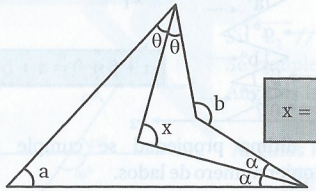
$\alpha + \beta = x + y$

4.



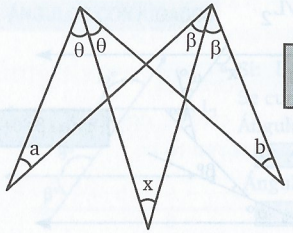
$\beta = \theta$

5. Ángulos determinados por dos bisectrices



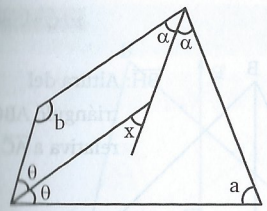
$x = \frac{a+b}{2}$

6.



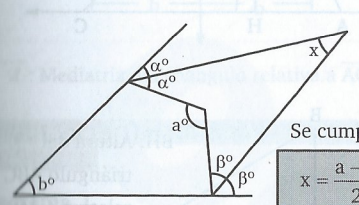
$x = \frac{a+b}{2}$

7.



$x = \frac{b-a}{2}$

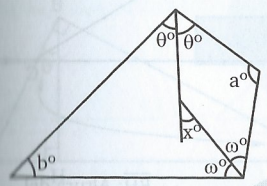
8.



Se cumple:

$x = \frac{a-b}{2}$

9.



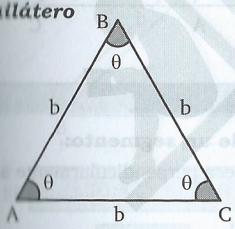
Se cumple

$x = \frac{a-b}{2}$

CLASIFICACIÓN DE TRIÁNGULOS

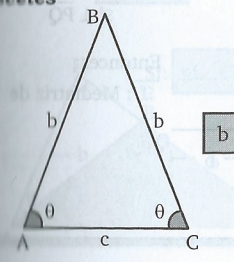
De acuerdo a la congruencia de sus lados

Equilátero



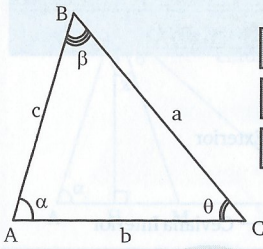
$\theta = 60°$

Isósceles



$b \neq c$ $\quad \theta < 90°$

Escaleno



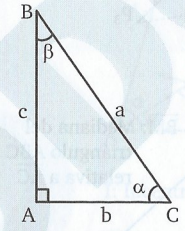
$a \neq b$ $\quad \alpha \neq \beta$

$b \neq c$ $\quad \beta \neq \theta$

$a \neq c$ $\quad \theta \neq \alpha$

De acuerdo a la congruencia de sus ángulos

Triángulo Rectángulo



Donde:

$\overline{BC}$: hipotenusa

$\overline{AB}$; $\overline{AC}$: catetos

Teorema:

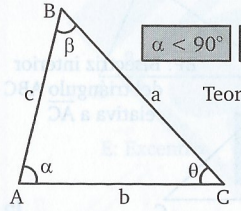
$\alpha + \beta = 90°$

$a^2 = b^2 + c^2$

Triángulo Oblicuángulo

(no tiene ángulo recto)

<u>Acutángulo</u>

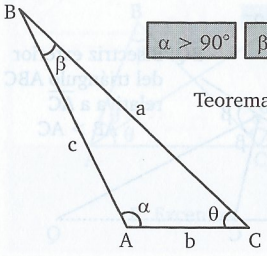


$\alpha < 90°$ $\quad \beta < 90°$ $\quad \theta < 90°$

Teorema:

$a^2 < b^2 + c^2$

<u>Obtusángulo</u>



$\alpha > 90°$ $\quad \beta < 90°$ $\quad \theta < 90°$

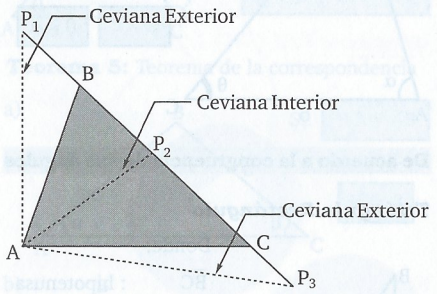
Teorema:

$a^2 > b^2 + c^2$

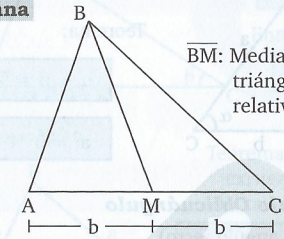
a: longitud del mayor lado

LÍNEAS NOTABLES ASOCIADAS AL TRIÁNGULO

CEVIANA

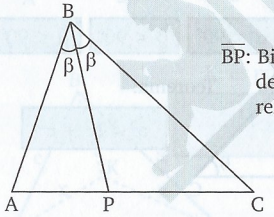


Mediana



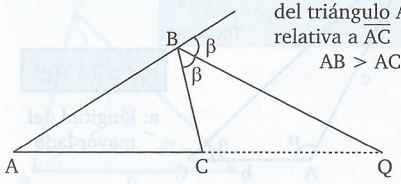
$\overline{BM}$: Mediana del triángulo ABC relativa a $\overline{AC}$

Bisectriz interior



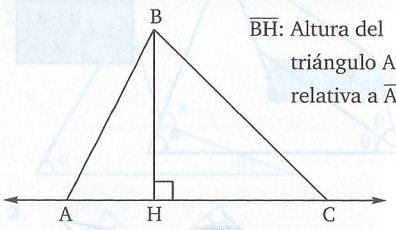
$\overline{BP}$: Bisectriz interior del triángulo ABC relativa a $\overline{AC}$

Bisectriz exterior

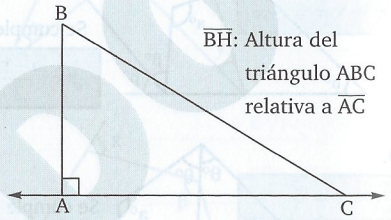


$\overline{BQ}$: Bisectriz exterior del triángulo ABC relativa a $\overline{AC}$
$AB > AC$

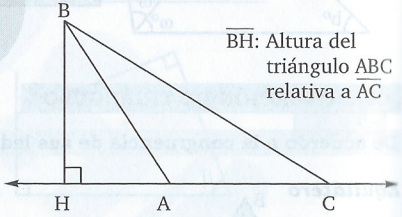
Altura



$\overline{BH}$: Altura del triángulo ABC relativa a $\overline{AC}$



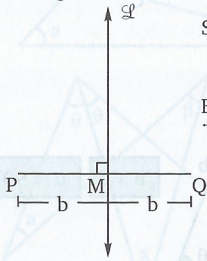
$\overline{BH}$: Altura del triángulo ABC relativa a $\overline{AC}$



$\overline{BH}$: Altura del triángulo ABC relativa a $\overline{AC}$

NOTA

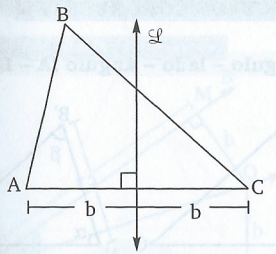
Mediatriz de un segmento:
Recta que biseca perpendicularmente a un segmento



Si: $PM = MQ$ y $\overline{\mathcal{L}} \perp \overline{PQ}$

Entonces:
$\overline{\mathcal{L}}$: Mediatriz de $\overline{PQ}$

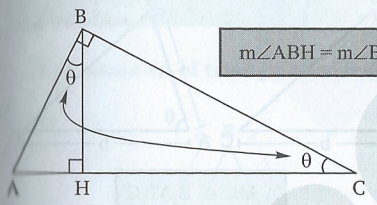
Mediatriz



$\overline{\mathcal{L}}$: Mediatriz del triángulo relativa a $\overline{AC}$

ÁNGULOS DETERMINADOS POR LÍNEAS NOTABLES

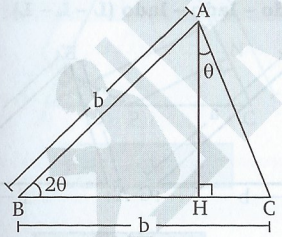
Teorema 1



$m\angle ABH = m\angle BCH$

Teorema 2

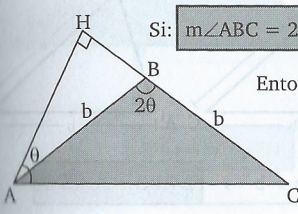
a)



Si: $m\angle ABC = 2(m\angle CAH)$

Entonces: $AB = BC$

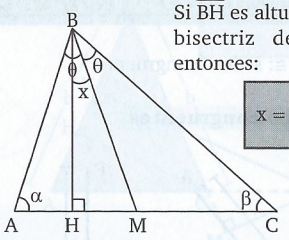
b)



Si: $m\angle ABC = 2(m\angle CAH)$

Entonces: $AB = BC$

Teorema 3

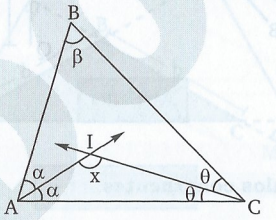


Si $\overline{BH}$ es altura y $\overline{BM}$ es bisectriz del $\angle ABC$, entonces:

$$x = \frac{\alpha - \beta}{2}$$

ÁNGULOS FORMADOS POR BISECTRICES DE LOS ÁNGULOS DE UN TRIÁNGULO

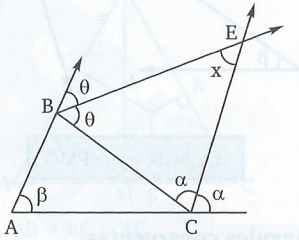
a)



I: Incentro

$$x = 90° + \frac{\beta}{2}$$

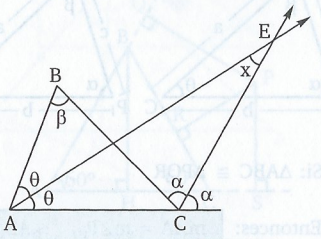
b)



E: Excentro

$$x = 90° - \frac{\beta}{2}$$

c)



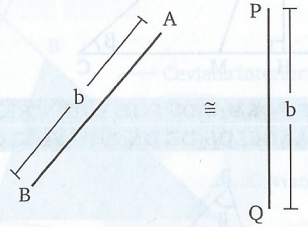
E: Excentro

$$x = \frac{\beta}{2}$$

CONGRUENCIA DE TRIÁNGULOS

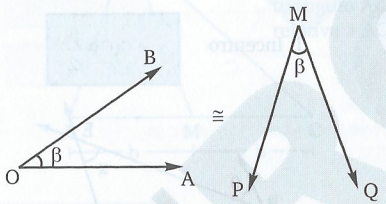
≅: se lee "congruente"

Segmentos congruentes



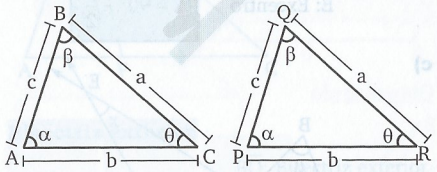
$\overline{AB} = \overline{PQ}$

Ángulos congruentes



$m\angle AOB = m\angle PMQ$

Triángulos congruentes



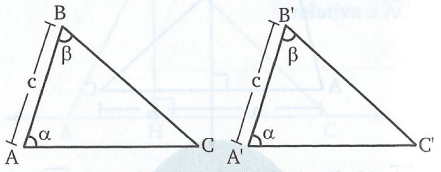
Si: $\Delta ABC \cong \Delta PQR$

Entonces:

$m\angle A = m\angle P$		$AB = PQ$
$m\angle B = m\angle Q$	y	$BC = QR$
$m\angle C = m\angle R$		$AC = PR$

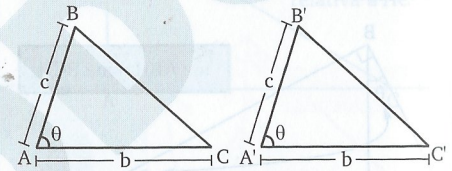
FORMAS DE RECONOCER A DOS TRIÁNGULOS CONGRUENTES

1. Ángulo – lado – ángulo (A – L – A)



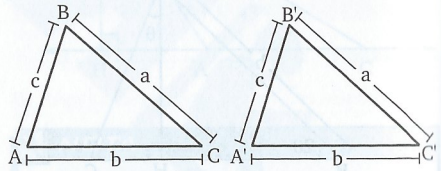
$\Delta ABC \cong \Delta A'B'C'$

2. Lado – ángulo – lado (L – A – L)



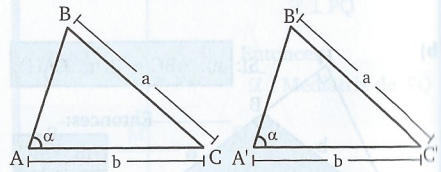
$\Delta ABC \cong \Delta A'B'C'$

3. Lado – lado – lado (L – L – L)



$\Delta ABC \cong \Delta A'B'C'$

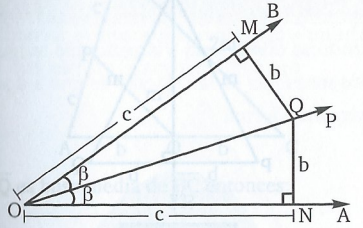
4. Lado – lado – Ángulo mayor ($L - L - A_{may.}$)



Si: a > b → $\Delta ABC \cong \Delta A'B'C'$

APLICACIONES DE LA CONGRUENCIA

TEOREMA DE LA BISECTRIZ



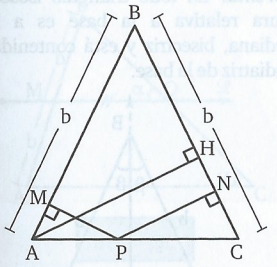
$\overrightarrow{OP}$: Bisectriz del $\angle AOB$

Entonces: $\overline{QN} = \overline{QM}$ ∧ $\overline{OM} = \overline{ON}$

NOTAS

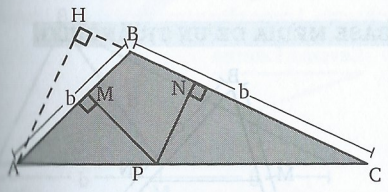
Teoremas en el triángulo isósceles

a)



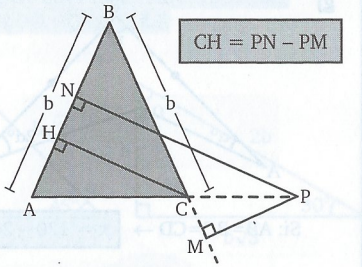
$\overline{AH} = \overline{PM} + \overline{PN}$

b)



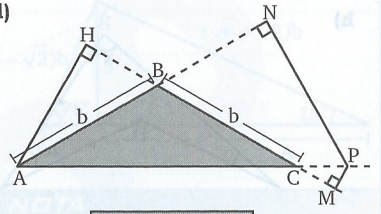
$\overline{AH} = \overline{PM} + \overline{PN}$

c)



$\overline{CH} = \overline{PN} - \overline{PM}$

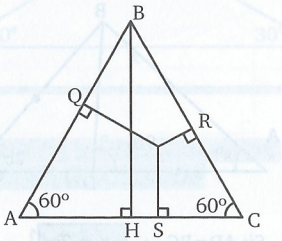
d)



$\overline{AH} = \overline{PN} - \overline{PM}$

Teoremas en el triángulo equilátero

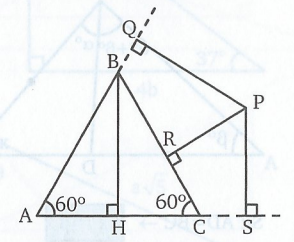
e)



Si: AB = BC = AC

→ $\overline{BH} = \overline{PQ} + \overline{PR} + \overline{PS}$

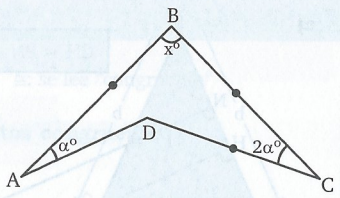
f)



Si: AB = BC = AC

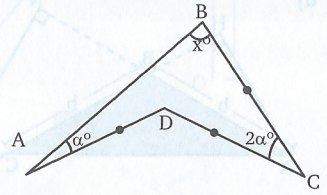
→ $\overline{BH} = \overline{PQ} + \overline{PS} - \overline{PR}$

g)



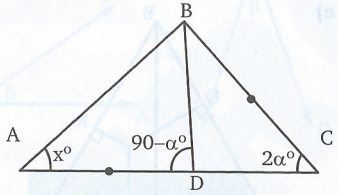
Si: AB=BC=CD → $x = 120 - 2\alpha$

h)



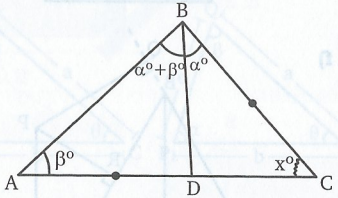
Si: BC=CD=AD → $x = 120 - \alpha$

i)



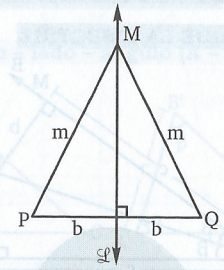
Si: AD=BC → $x = 2\alpha$

j)



Si: AD=BC → $x = \beta$

TEOREMA DE LA MEDIATRIZ



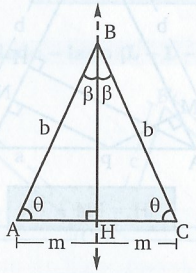
$\overleftrightarrow{\mathcal{L}}$: mediatriz de $\overline{PQ}$

M: punto de la mediatriz

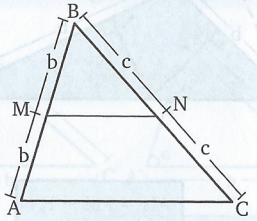
Entonces: $MP = MQ$

NOTA

Teorema: En todo triángulo isósceles; la altura relativa a la base es a su vez mediana, bisectriz y está contenida en la mediatriz de la base.

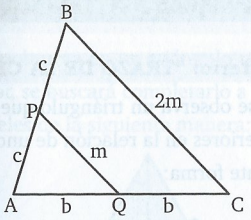


BASE MEDIA DE UN TRIÁNGULO



$\overline{MN}$: base media del $\triangle ABC$

TEOREMA DE LA BASE MEDIA



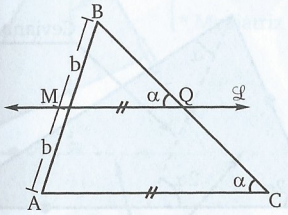
Si $\overline{PQ}$ es base media de $\overline{BC}$ entonces:

→ $\overline{PQ} \,//\, \overline{BC}$

→ $PQ = \dfrac{BC}{2}$

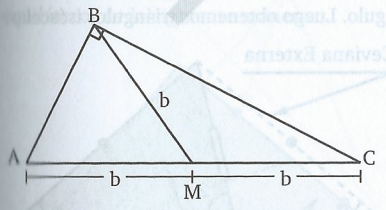
NOTA

Si $BM = MA$ y $\overleftrightarrow{\mathcal{L}} \,//\, \overline{AC}$ entonces:



$BQ = QC$

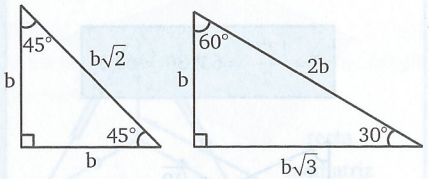
TEOREMA DE LA MEDIANA RELATIVA A LA HIPOTENUSA

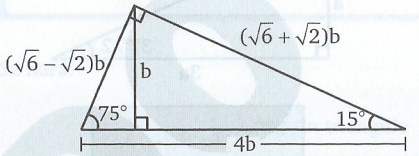


Si $\overline{BM}$ es mediana entonces:

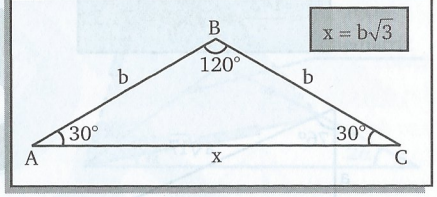
$BM = \dfrac{AC}{2}$

TRIÁNGULOS RECTÁNGULOS DE ÁNGULOS NOTABLES

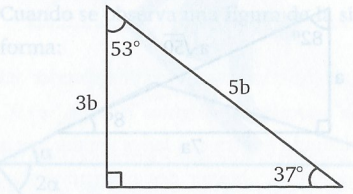


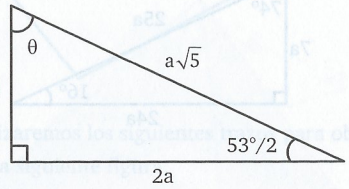


NOTA



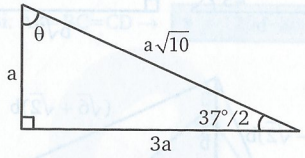
TRIÁNGULOS RECTÁNGULOS DE ÁNGULOS APROXIMADOS





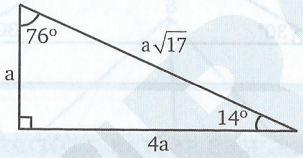
$$\frac{53°}{2} = 26°30' = 26,5°$$

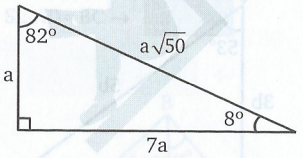
$$\theta = \frac{127°}{2} = 63°30' = 63,5°$$

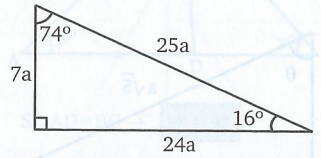


$$\frac{37°}{2} = 18°30' = 18,5°$$

$$\theta = \frac{143°}{2} = 71°30' = 71,5°$$



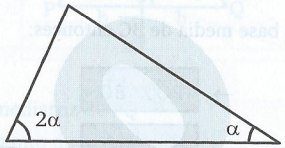




CRITERIOS PARA HACER TRAZOS AUXILIARES EN TRIÁNGULOS

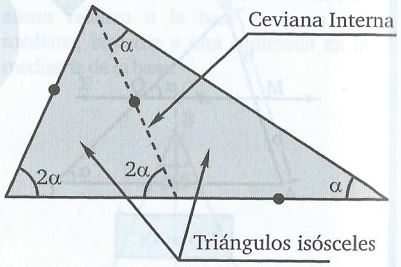
1er. Criterio: "TRAZO DE LA CEVIANA"

Cuando se observa un triángulo que tenga ángulos interiores en la relación de uno a dos, de la siguiente forma:



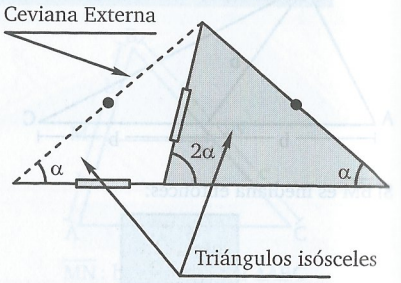
TRAZANDO CEVIANA INTERIOR

Se trazará una ceviana interna tal que forme un ángulo igual al mayor en la misma base del triángulo. Luego obtenemos triángulos isósceles.



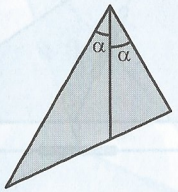
TRAZANDO CEVIANA EXTERIOR

Se trazará una ceviana externa tal que forme un ángulo igual al menor en la misma base del triángulo. Luego obtenemos triángulos isósceles.

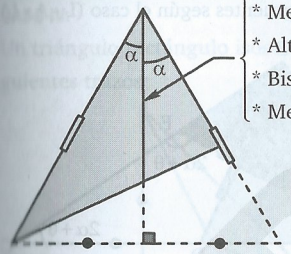


2do. Criterio: "COMPLETANDO A UN TRIÁNGULO ISÓSCELES"

Cuando se observa en un triángulo una bisectriz interior, se buscará completarlo a un triángulo isósceles de la siguiente manera:



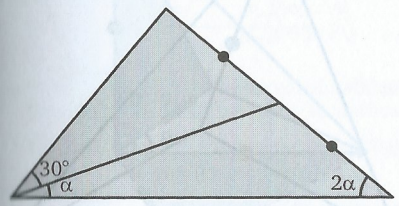
Se hará:



* Mediana
* Altura
* Bisectriz interior
* Mediatriz

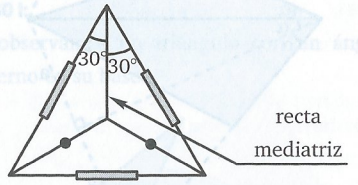
3er. Criterio: "COMPLETANDO UN TRIÁNGULO EQUILÁTERO"

Cuando se observa en un triángulo un ángulo de 30° y como éste valor es la mitad de 60°, se buscará formar externamente un triángulo equilátero, de la siguiente manera:

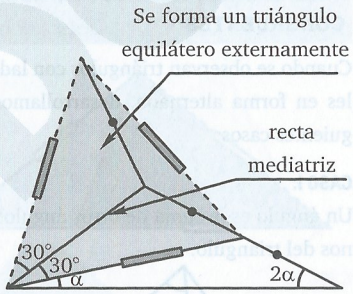


OBSERVACIÓN

Tener presente en un triángulo equilátero

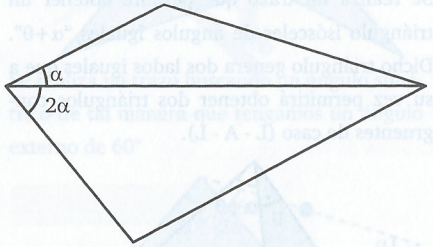


Se hará:

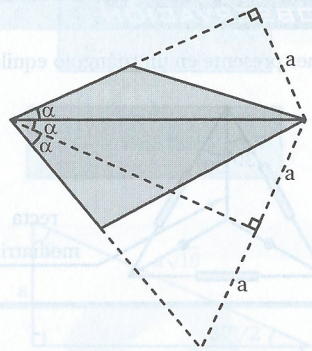


4to. Criterio: "PROPIEDAD DE LA BISECTRIZ"

Cuando se observa una figura de la siguiente forma:



Realizaremos los siguientes trazos para obtener la siguiente figura.

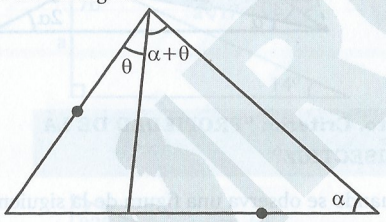


5to. Criterio: "BUSCANDO TRIÁNGULOS CONGRUENTES"

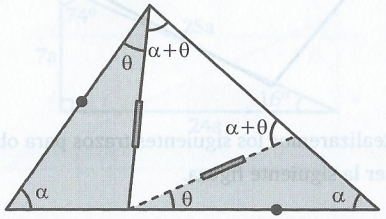
Cuando se observan triángulos con lados iguales en forma alternada, desarrollamos los siguientes casos:

CASO I:

Un ángulo es la suma de otros ángulos internos del triángulo.

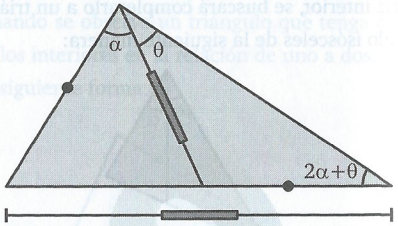


Se realiza un trazo que permite obtener un triángulo isósceles de ángulos iguales "α+θ". Dicho triángulo genera dos lados iguales que a su vez permitirá obtener dos triángulos congruentes de caso (L - A - L).

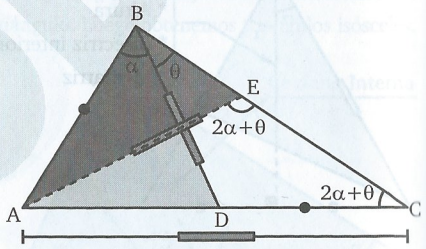


CASO II:

Cuando una ceviana es de la misma longitud que el lado mayor del triángulo.



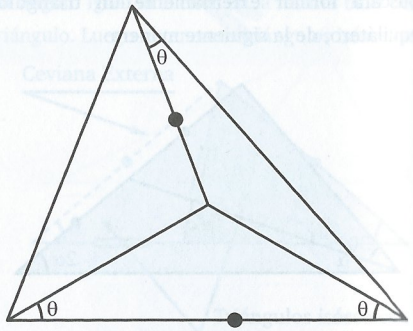
Se realiza el trazo de una ceviana para obtener triángulos isósceles; el cual genera dos lados iguales y dos ángulos iguales. Generando dos triángulos congruentes según el caso (L - A - L)



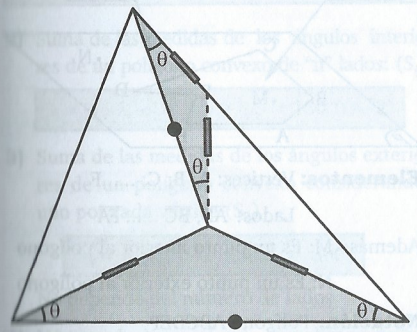
$\Delta ABE \cong \Delta ABD$

CASO III:

Si tenemos la siguiente figura con las siguientes características

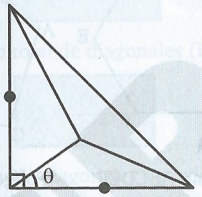


Se realiza el trazo de una ceviana para obtener triángulos isósceles; el cual genera dos triángulos congruentes.

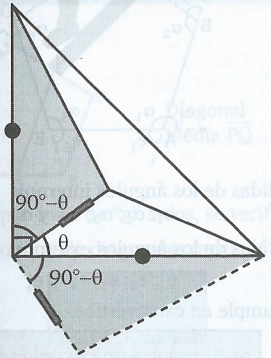


CASO IV:

Un triángulo rectángulo isósceles con los siguientes trazos:



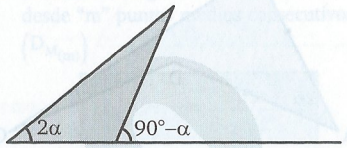
Se realiza un trazo buscando dos lados iguales con ángulos iguales para obtener dos triángulos congruentes del caso (L - A - L)



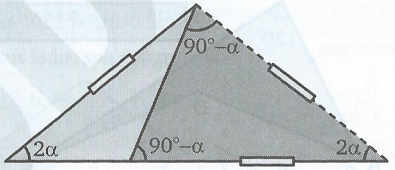
6to. Criterio: "TRAZO DEL ÁNGULO SIMÉTRICO (ESPEJO)"

CASO I:

Si observamos un triángulo con un ángulo externo en su base.

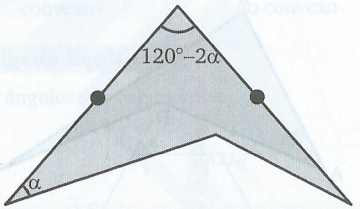


Se realiza un trazo buscando completar dos triángulo isósceles.

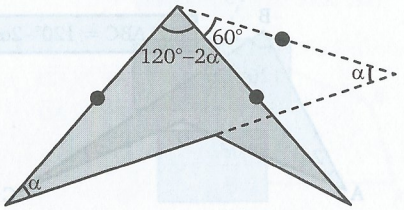


CASO II:

Si tenemos está figura con las siguientes características.



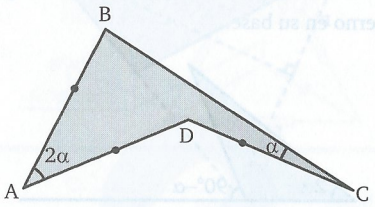
Se realiza un trazo buscando un ángulo simétrico de tal manera que tengamos un ángulo externo de 60°



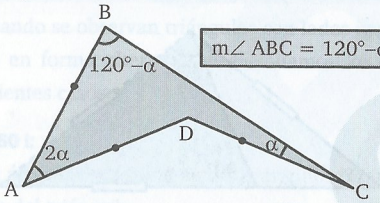
7mo. Criterio: "BUSCANDO CUADRILÁTEROS CÓNCAVOS"

CASO I:

Si tenemos:



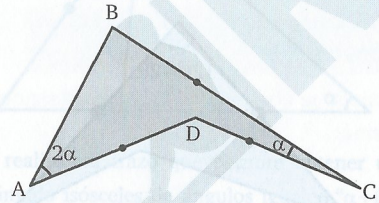
Se realiza un trazo buscando completar dos triángulo isósceles.



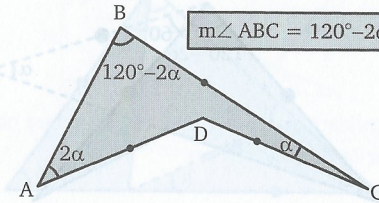
$m\angle ABC = 120° - \alpha$

CASO II:

Si tenemos está figura con las siguientes características.

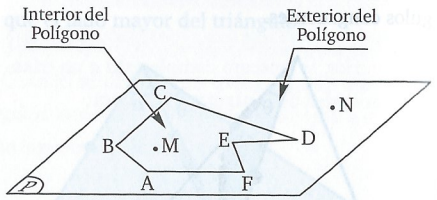


Se realiza un trazo buscando un ángulo simétrico de tal manera que tengamos un ángulo externo de 60°



$m\angle ABC = 120° - 2\alpha$

POLÍGONO



Elementos: Vértices: A; B; C; .....F

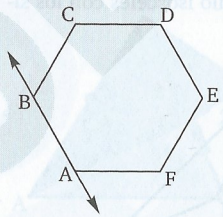
Lados: $\overline{AB}$, $\overline{BC}$ ..... $\overline{FA}$

Además: M: Es un punto interior al polígono

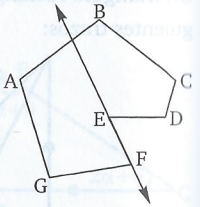
N: Es un punto exterior al polígono

Notación: Polígono ABCDEF

POLÍGONO CONVEXO Y POLÍGONO NO CONVEXO

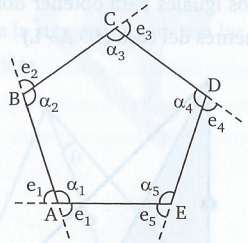


Polígono ABCDEF es convexo



Polígono ABCDEFG es no convexo

ÁNGULOS DEL POLÍGONO CONVEXO



Medidas de los ángulos internos:

α_1; α_2; α_3; α_4 y α_5

Medidas de los ángulos externos:

e_1; e_2; e_3; e_4 y e_5

Se cumple en cada vértice:

$m\angle Interior + m\angle Exterior = 180°$

PROPIEDADES DE LOS POLÍGONOS

1) En todo polígono convexo de "n" lados.

$\#vértices = \#ángulos\ internos = \#lados = n$

2) Suma de las medidas de los ángulos interiores de un polígono convexo de "n" lados: (S_i)

$S_i = \alpha_1 + \alpha_2 + ... + \alpha_n = 180°(n-2)$

3) Suma de las medidas de los ángulos exteriores de un polígono convexo considerando uno por cada vértice (S_e)

$S_e = e_1 + e_2 + ... + e_n = 360°$

No depende del número de lados

4) Número de diagonales de un polígono de "n" lados:

- Número de diagonales trazadas desde un vértice $(D_{(1v)})$

$$D_{(1v)} = n - 3$$

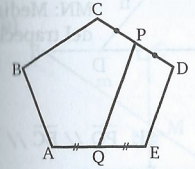
- Número total de diagonales (D)

$$D = \frac{n(n-3)}{2}$$

- Número de diagonales trazadas desde "m" vértices consecutivos $(D_{(m)})$

$$D_{(m)} = mn - \frac{(m+1)(m+2)}{2}$$

NOTA



Diagonal media $\overline{PQ}$

En todo polígono de "n" lados, se verifica:

- Número total de diagonales medias (D_M)

$$D_M = \frac{n(n-1)}{2}$$

- Número de diagonales medias trazadas desde el punto medio de uno de sus lados $(D_{M_{(1L)}})$

$$D_{M_{(1.L)}} = n - 1$$

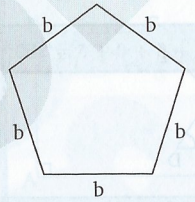
- Número de diagonales medias trazadas desde "m" puntos medios consecutivos $(D_{M_{(m)}})$

$$D_{M_{(m)}} = nm - \frac{m(m+1)}{2}$$

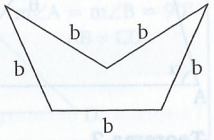
POLÍGONOS ESPECIALES

Polígono Equilátero

(sus lados son congruentes)



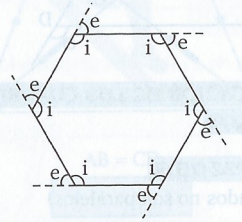
Polígono equilátero convexo



Polígono equilátero no convexo

Polígono Equiángulo

(sus ángulos son congruentes)

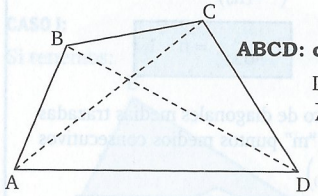


$$i = \frac{180°(n-2)}{n}$$

$$e = \frac{360°}{n}$$

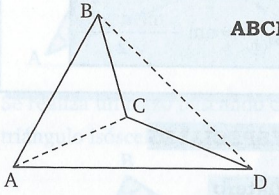
CUADRILÁTERO

(Polígono de cuatro lados)



ABCD: convexo

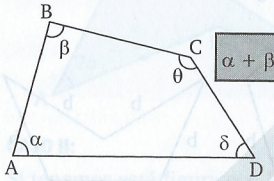
Diagonales: $\overline{AC}$ y $\overline{BD}$



ABCD: no convexo o cóncavo

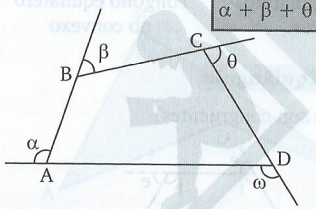
Diagonales: $\overline{AC}$ y $\overline{BD}$

Teorema 1



$$\alpha + \beta + \theta + \delta = 360°$$

Teorema 2

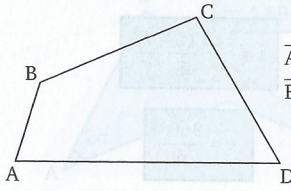


$$\alpha + \beta + \theta + \omega = 360°$$

CLASIFICACIÓN DE LOS CUADRILÁTEROS CONVEXOS

A) TRAPEZOIDE

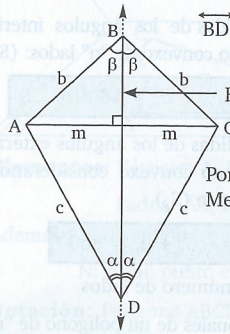
(sus lados no son paralelos)



$\overline{AB} \not\parallel \overline{CD}$ y $\overline{BC} \not\parallel \overline{AD}$

NOTA

Trapezoide Simétrico. Trapezoide donde una de las diagonales es parte de la mediatriz de la otra.



$\overleftrightarrow{BD}$: Mediatriz de $\overline{AC}$

Eje de simetría

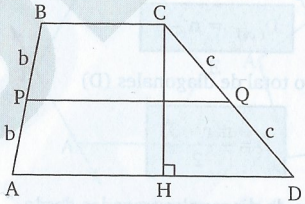
Por propiedad de la Mediatriz

$AB = BC$

$AD = CD$

B) TRAPECIO

(tiene dos lados paralelos llamados bases)



$\overline{BC} \parallel \overline{AD}$ y $\overline{AB} \not\parallel \overline{CD}$

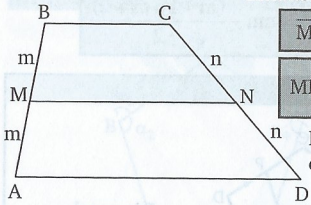
$\overline{CH}$: Altura

$\overline{PQ}$: Mediana del trapecio

Teorema 1

Si: $\overline{BC} \parallel \overline{AD}$

Entonces:



$\overline{MN} \parallel \overline{BC}$ y

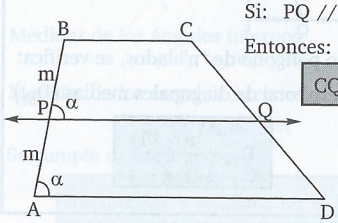
$MN = \dfrac{BC + AD}{2}$

MN: Mediana del trapecio

Teorema 2

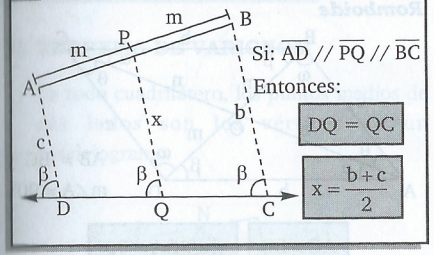
Si: $\overline{PQ} \parallel \overline{BC} \parallel \overline{AD}$

Entonces:



$CQ = QD$

NOTA



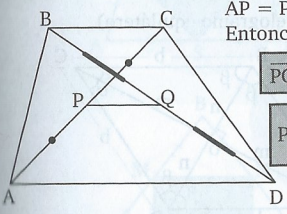
Si: $\overline{AD} \parallel \overline{PQ} \parallel \overline{BC}$

Entonces:

$DQ = QC$

$x = \dfrac{b + c}{2}$

Teorema 3



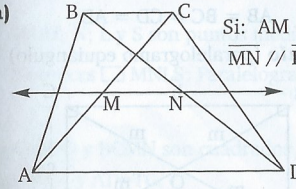
Si: $\overline{BC} \parallel \overline{AD}$;
$AP = PC$ y $BQ = QD$
Entonces:

$\overline{PQ} \parallel \overline{BC} \parallel \overline{AD}$

$PQ = \dfrac{AD - BC}{2}$

NOTA

a)

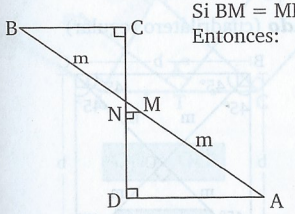


Si: $AM = MC$ y $\overline{MN} \parallel \overline{BC} \parallel \overline{AD}$

Entonces:

$BN = ND$ ∧ $NM = \dfrac{AD - BC}{2}$

b)

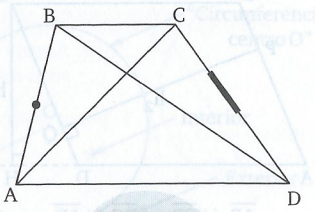


Si $BM = MD$
Entonces:

$CN = ND$ ∧ $NM = \dfrac{AD - BC}{2}$

CLASES DE TRAPECIO

Trapecio escaleno



$AB \neq CD$

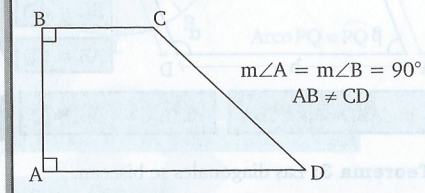
$m\angle A \neq m\angle D$

$m\angle B \neq m\angle C$

$AC \neq BD$

NOTA

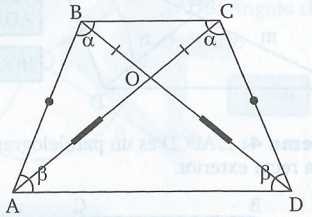
Trapecio rectángulo



$m\angle A = m\angle B = 90°$

$AB \neq CD$

Trapecio isósceles



$AB = CD$

$m\angle A = m\angle D$

$m\angle B = m\angle C$

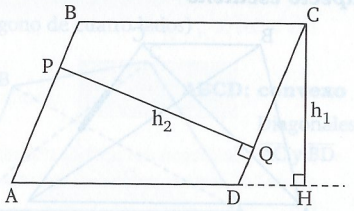
$AC = BD$

$AO = OD$

$BO = OC$

C) PARALELOGRAMO

(sus lados son paralelos)

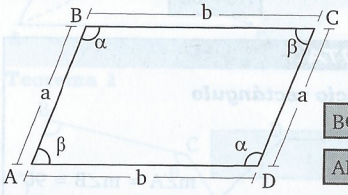


$\overline{AB} \,//\, \overline{CD}$ y $\overline{BC} \,//\, \overline{AD}$

$\overline{CH}$: altura relativa a $\overline{AD}$

$\overline{PQ}$: altura relativa a $\overline{CD}$

Teorema: Los lados y ángulos opuestos son congruentes



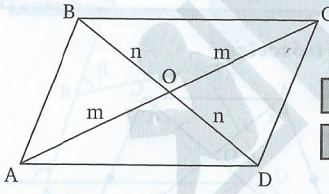
BC = AD

AB = CD

m∠ABC = m∠CDA

m∠BAD = m∠BCD

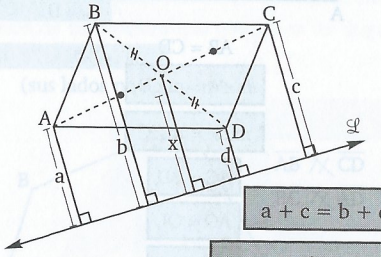
Teorema 3: Las diagonales se bisecan.



AO = OC

BO = OD

Teorema 4: ▱ABCD es un paralelogramo y $\mathcal{L}$ es una recta exterior.

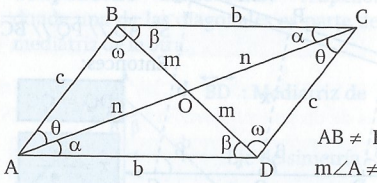


a + c = b + d

4x = a + b + c + d

CLASES DE PARALELOGRAMO

Romboide



AB ≠ BC y m∠A ≠ 90°

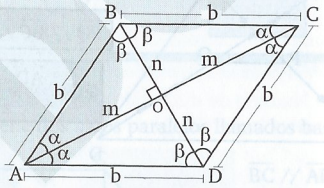
AC ≠ BD

m∠BOA ≠ 90°

m∠BAO ≠ m∠DAO

m∠ABO ≠ m∠CBO

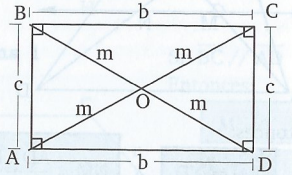
Rombo (paralelogramo equilátero)



$\overline{AC} \perp \overline{BD}$

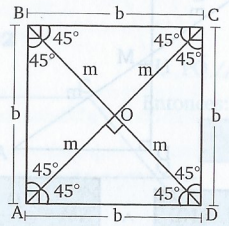
AB = BC = CD = AD

Rectángulo (paralelogramo equiángulo)



AC = BD

Cuadrado (cuadrilátero regular)

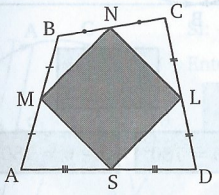


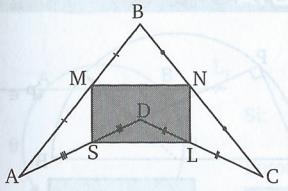
AC = BD $\overline{AC} \perp \overline{BD}$

TEOREMAS ADICIONALES

1) TEOREMA DE VARIGNOM

En todo cuadrilátero, los puntos medios de sus lados son los vértices de un paralelogramo.

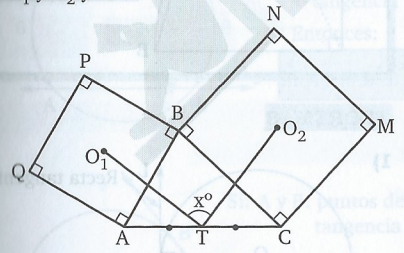




Si: M; N; L y S son puntos medios

Entonces □ MNLS: Paralelogramo

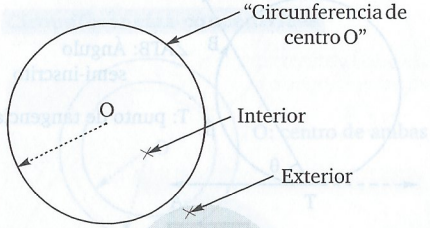
2) Si ABPQ y BCMN son cuadrados de centros O_1 y O_2 y AT=TC.



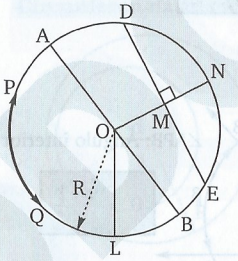
$O_1T = O_2T$

x = 90

CIRCUNFERENCIA



Elementos:



Radio: $\overline{OL}$ (OL = R)

Cuerda: $\overline{DE}$

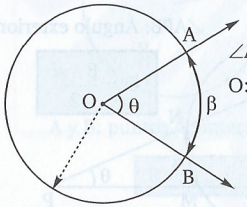
Flecha o sagita: $\overline{MN}$

Diámetro: $\overline{AB}$ (AB = 2R)

Arco $PQ \equiv \widehat{PQ}$

POSICIONES RELATIVAS ENTRE UN ÁNGULO Y UNA CIRCUNFERENCIA

Ángulo Central

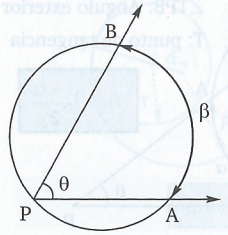


∠AOB: Ángulo central

O: centro

β = θ

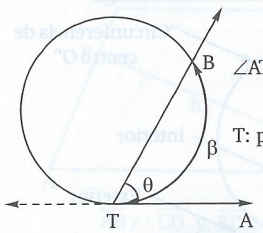
Ángulo Inscrito



∠APB: Ángulo inscrito

$\theta = \frac{\beta}{2}$

Ángulo Semi-inscrito

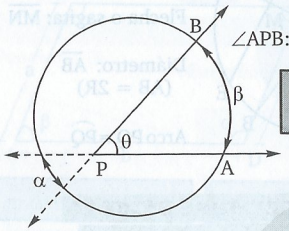


∠ATB: Ángulo semi-inscrito

T: punto de tangencia

$$\theta = \frac{\beta}{2}$$

Ángulo Interior

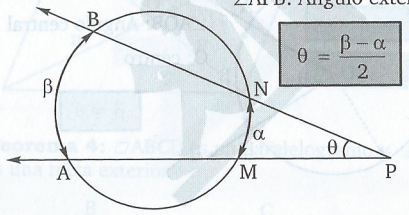


∠APB: Ángulo interior

$$\theta = \frac{\alpha+\beta}{2}$$

Ángulo Exterior

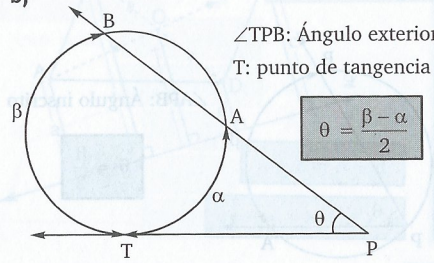
a)



∠APB: Ángulo exterior

$$\theta = \frac{\beta-\alpha}{2}$$

b)

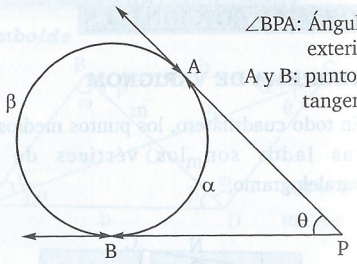


∠TPB: Ángulo exterior

T: punto de tangencia

$$\theta = \frac{\beta-\alpha}{2}$$

c)



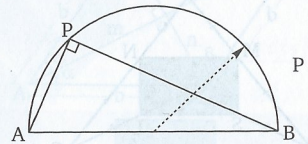
∠BPA: Ángulo exterior

A y B: puntos de tangencia

$$\theta = \frac{\beta-\alpha}{2}$$

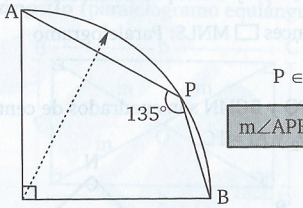
$\alpha + \theta = 180°$ $\quad$ $\beta - \theta = 180°$

NOTA



$P \in \widehat{AB}$

$m\angle APB = 90°$

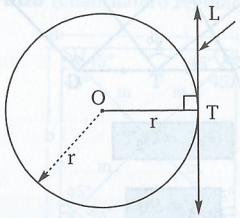


$P \in \widehat{AB}$

$m\angle APB = 135°$

TEOREMAS

1)

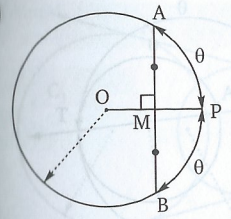


Recta tangente

T: punto de tangencia

$\overline{OT} \perp \overleftrightarrow{L}$ $\quad$ $OT = r$

2)



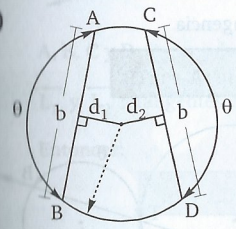
Si: $\overline{OP} \perp \overline{AB}$

Entonces:

$AM = MB$

$m\widehat{AP} = m\widehat{PB}$

3)



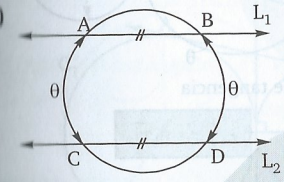
Si: $AB = CD$

Entonces:

$m\widehat{AB} = m\widehat{CD}$

$d_1 = d_2$

4)

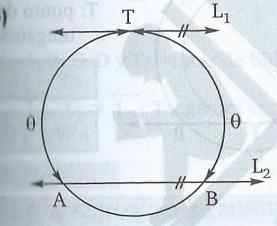


Si: $\overleftrightarrow{L_1} // \overleftrightarrow{L_2}$

Entonces:

$m\widehat{AC} = m\widehat{BD}$

5)



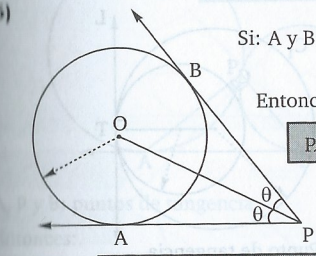
Si: $\overleftrightarrow{L_1} // \overleftrightarrow{L_2}$

T: punto de tangencia

Entonces:

$m\widehat{AT} = m\widehat{TB}$

6)



Si: A y B: puntos de tangencia

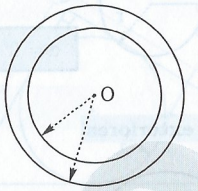
Entonces:

$PA = PB$

Además: $\overrightarrow{PO}$: Bisectriz del ángulo APB

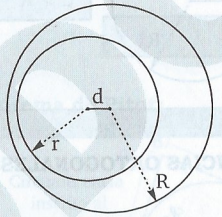
POSICIONES RELATIVAS ENTRE DOS CIRCUNFERENCIAS

Circunferencias concéntricas



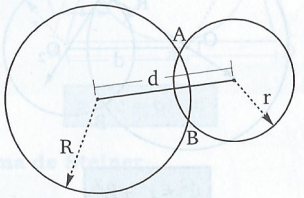
O: centro de ambas

Circunferencias excéntricas



$d < R - r$

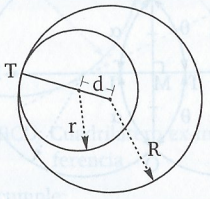
Circunferencias secantes



A y B: puntos de intersección

$R - r < d < R + r$

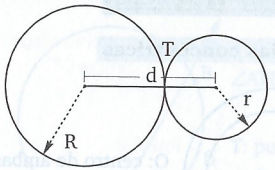
Circunferencias tangentes interiores



T: Punto de tangencia

$d = R - r$

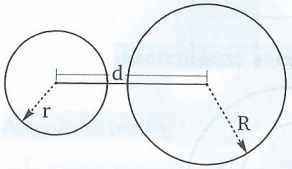
Circunferencias tangentes exteriores



T: Punto de tangencia

$d = R + r$

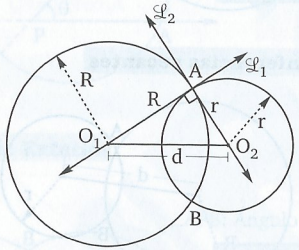
Circunferencias exteriores



$d > R + r$

NOTA

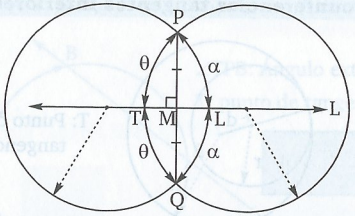
CIRCUNFERENCIAS ORTOGONALES



$d^2 = r^2 + R^2$

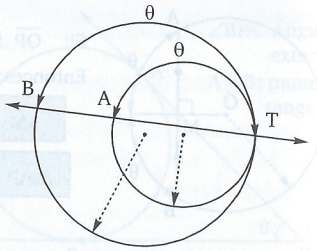
TEOREMAS

1) $\overline{PQ}$: Cuerda común



$PM = MQ$ | $m\widehat{PT} = m\widehat{TQ}$ | $m\widehat{PL} = m\widehat{LQ}$

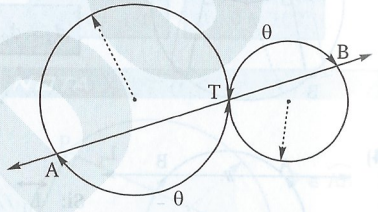
2)



T: punto de tangencia

$m\widehat{TA} = m\widehat{TB}$

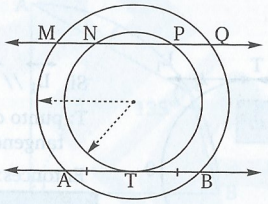
3)



T: punto de tangencia

$m\widehat{TA} = m\widehat{TB}$

4)



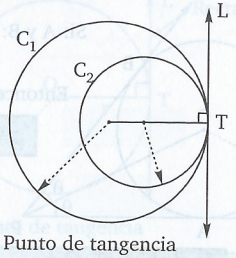
T: punto de tangencia

$AT = TB$

$MN = PQ$

RECTAS TANGENTES COMUNES A DOS CIRCUNFERENCIAS

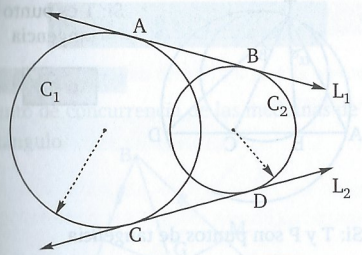
a)



T : Punto de tangencia

$\overleftrightarrow{L}$: Tangente común a C_1 y C_2

b)



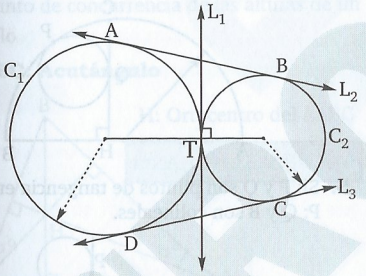
A, B, C y D: Puntos de tangencia

$\overleftrightarrow{L_1}$ y $\overleftrightarrow{L_2}$: Tangentes comunes a C_1 y C_2

Entonces:

$AB = CD$

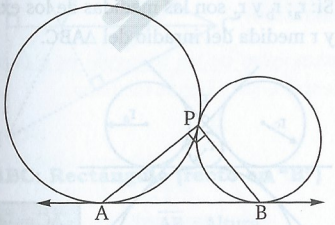
c)



A, B, C, D y T: Puntos de tangencia.

$\overleftrightarrow{L_1}$, $\overleftrightarrow{L_2}$ y $\overleftrightarrow{L_3}$: Tangentes comunes a C_1 y C_2

PROPIEDAD



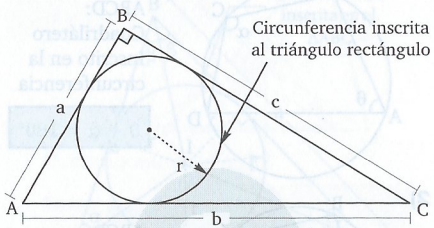
A, P y B: puntos de tangencia

Entonces:

$m\angle APB = 90°$

POSICIONES RELATIVAS ENTRE UN POLÍGONO Y UNA CIRCUNFERENCIA

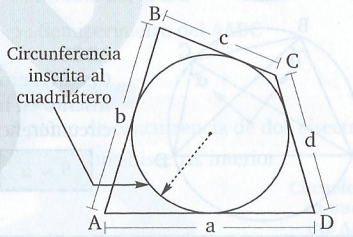
Teorema de Poncelet



r: inradio del ΔABC

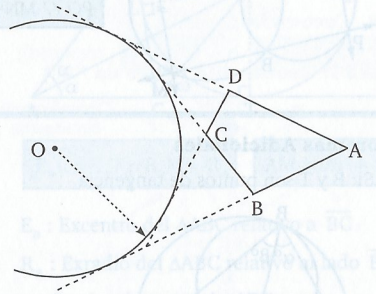
$a + c = b + 2r$

Teorema de Pitot:



$a + c = b + d$

Teorema de Steiner



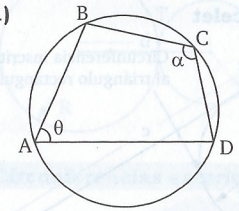
$\square ABCD$: Cuadrilátero ex-inscrito a la circunferencia

Se cumple:

$AB - CD = AD - BC$

PROPIEDADES DEL CUADRILÁTERO INSCRITO EN UNA CIRCUNFERENCIA

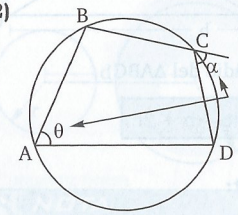
1)



ABCD: Cuadrilátero Inscrito en la circunferencia

$\theta + \alpha = 180°$

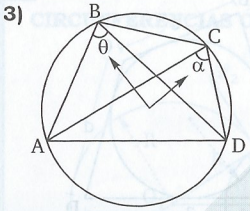
2)



ABCD: Cuadrilátero Inscrito en la circunferencia

$\theta = \alpha$

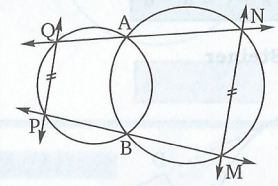
3)



ABCD: Cuadrilátero Inscrito en la circunferencia

$\theta = \alpha$

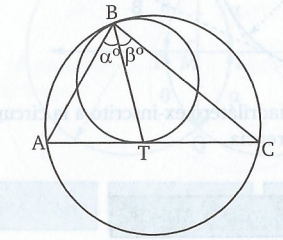
NOTA



$\overline{PQ} \,//\, \overline{MN}$

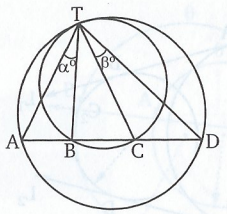
Teoremas Adicionales

1. Si: B y T son puntos de tangencia



$\alpha = \beta$

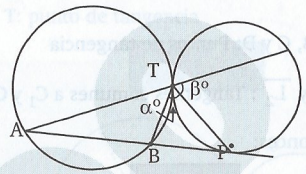
2.



Si: T es punto de tangencia

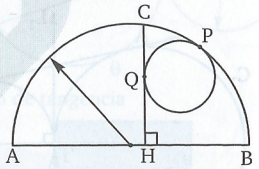
$\alpha = \beta$

3. Si: T y P son puntos de tangencia

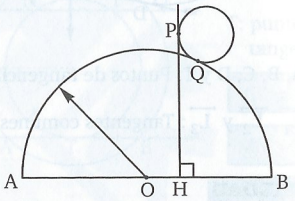


$\alpha = \beta$

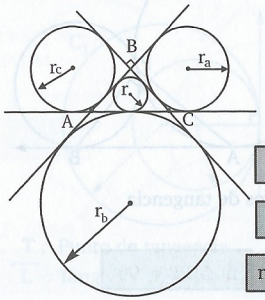
4. Si: P y Q son puntos de tangencia entonces A; Q y P son colineales.



5. Si: P y Q son puntos de tangencia entonces P; Q y B son colineales.



6. Si: r_a; r_b y r_c son las medidas de los exradios y r medida del inradio del ΔABC.



$AC = r_a + r_c$

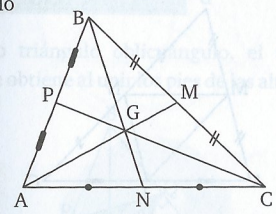
$AC = r_b - r$

$r_b = r_a + r_c + r$

PUNTOS NOTABLES

BARICENTRO

Punto de concurrencia de las medianas de un triángulo



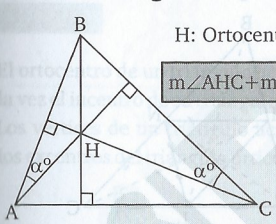
G: Baricentro del ΔABC

$AG = 2(GM)$ $BG = 2(GN)$ $CG = 2(GP)$

ORTOCENTRO:

Es el punto de concurrencia de las alturas de un triángulo

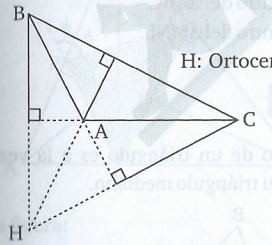
- **ΔABC: Acutángulo**



H: Ortocentro del ΔABC

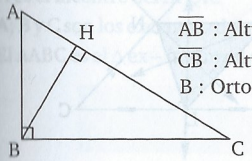
$m\angle AHC + m\angle ABC = 180°$

- **ΔABC: Obtusángulo (obtuso en "A")**



H: Ortocentro del ΔABC

- **ΔABC: Rectángulo (recto en "B")**



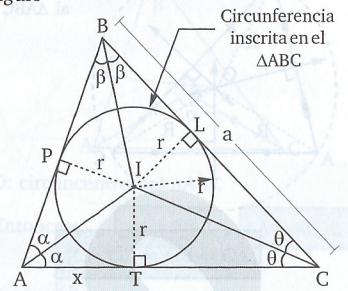
$\overline{AB}$: Altura

$\overline{CB}$: Altura

B : Ortocentro del ΔABC

INCENTRO:

Punto de concurrencia de las bisectrices de un triángulo



$IL = IT = IP = r$ $x = p - a$

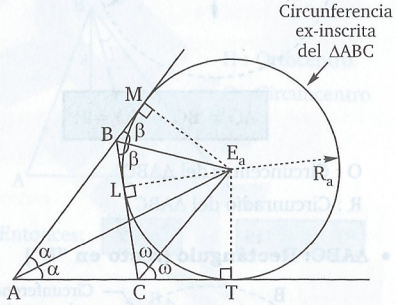
I : incentro del Δ ABC

r : Inradio del ΔABC

p : Semiperímetro del ΔABC

EX – CENTRO:

Es el punto de concurrencia de dos bisectrices exteriores y una bisectriz interior



$E_aL = E_aT = E_aM = R_a$ $AM = AT = p$

E_a : Excentro del ΔABC relativo a $\overline{BC}$

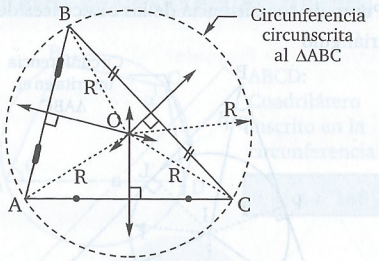
R_a : Exradio del ΔABC relativo al lado $\overline{BC}$

p : Semiperímetro del ΔABC

CIRCUNCENTRO:

Punto de concurrencia de las mediatrices de los lados de un triángulo

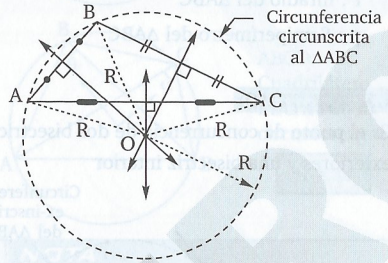
- **ΔABC: Acutángulo**



$AO = BO = CO = R$

O : Circuncentro del ΔABC
R : Circunradio del ΔABC

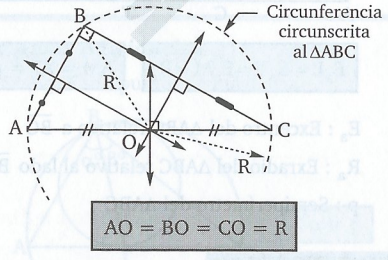
- **ΔABC: Obtusángulo**



$AO = BO = CO = R$

O : Circuncentro del ΔABC
R : Circunradio del ΔABC

- **ΔABC: Rectángulo (recto en "B")**

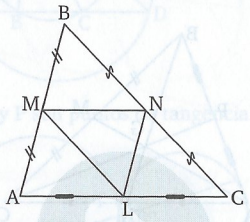


$AO = BO = CO = R$

O : Circuncentro del ΔABC
O : Punto medio de la hipotenusa
R : Circunradio del ΔABC

TRIÁNGULO MEDIANO

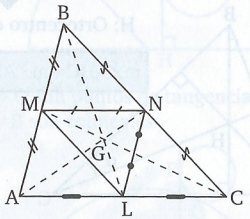
Se obtiene al unir los puntos medios de los lados de un triángulo.



Si: M; N y L son puntos medios → Δ MNL es el triángulo mediano o complementario y el triángulo ABC se llama antimediano o anticomplementario.

TEOREMA

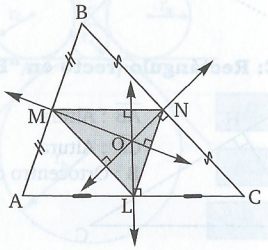
El baricentro de un triángulo es a la vez el baricentro de su triángulo mediano.



En la figura:
* G es baricentro del ΔABC.
* G es baricentro del ΔMNL.

TEOREMA

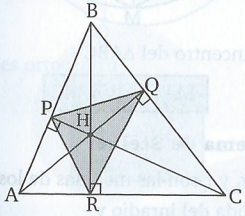
El circuncentro de un triángulo es a la vez el ortocentro se su triángulo mediano.



En la figura:
* O es circuncentro del ΔABC.
* O es ortocentro del ΔMNL.

TRIÁNGULO ÓRTICO

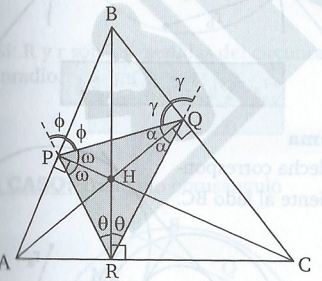
En todo triángulo oblicuángulo, el triángulo órtico se obtiene al unir los pies de las alturas.



En la figura:
* ΔPQR es el triángulo órtico.
* ΔABC es el triángulo antiórtico.

TEOREMA

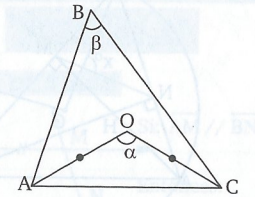
* El ortocentro de un triángulo acutángulo es a la vez el incentro de su triángulo órtico.
* Los vértices de un triángulo acutángulo son los excentros del triángulo órtico.



En la figura:
* H es el ortocentro del ΔABC.
* H es el incentro del ΔPQR.
* A; B y C son los excentros del ΔPQR.
* El ΔABC es el Δ ex – incentral del Δ PQR.

TEOREMAS

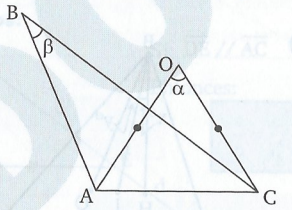
1)



O: circuncentro del ΔABC

Entonces: $\alpha = 2\beta$ $AO = OC$

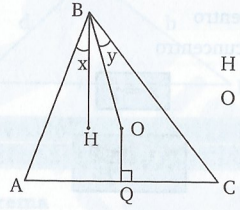
2)



O: circuncentro del ΔABC (obtuso en A)

Entonces: $\alpha = 2\beta$ $AO = OC$

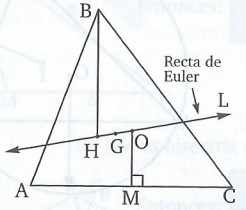
3)



H : Ortocentro
O : Circuncentro

Entonces: $x = y$ $BH = 2(OQ)$

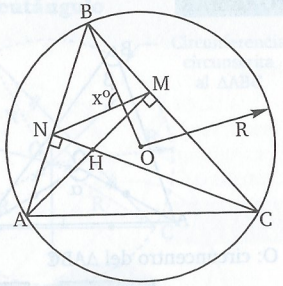
4)



H : ortocentro
G : baricentro
O : Cincuncentro
Se cumple: H, G y O son colineales

$HG = 2(GO)$
$BH = 2(OM)$

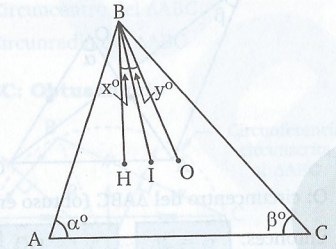
5)



Si: H: Ortocentro
O: Circuncentro

$x = 90°$

6)

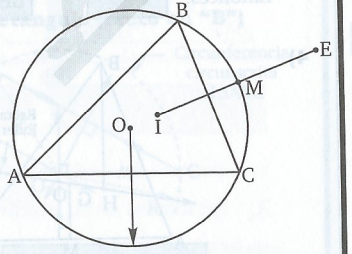


Si: H: Ortocentro
I : Incentro
O: Circuncentro

$x = y$

$$x = \frac{\alpha - \beta}{2}$$

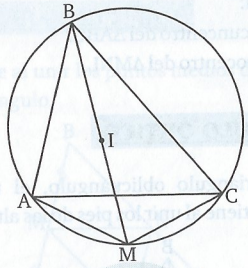
7)



Si: I : Incentro
E : Excentro

$IM = ME$

8.

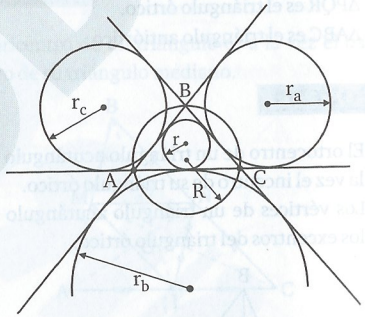


Si: I : Incentro del ΔABC

$MA = MI = MC$

9. Teorema de Steiner

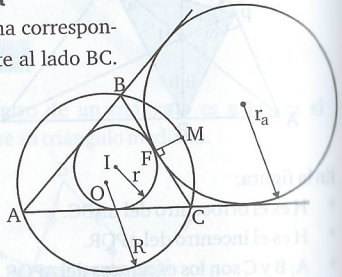
Si: r_a; r_b y r_c son las medidas de los exradios.
r: medida del inradio y
R: medida del circunradio



$$r_a + r_b + r_c = 4R + r$$

Teorema

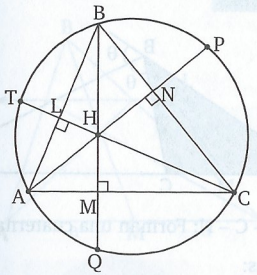
$\overline{MF}$: Flecha correspondiente al lado BC.



Si: $\overset{\frown}{BM} = m\overset{\frown}{MC}$ y $\overline{MF} \perp \overline{BC}$

$$MF = \frac{r_a - r}{2}$$

Teorema

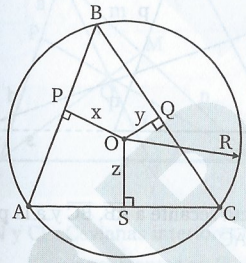


Si: H es ortocentro

$HM = MQ$
$HN = NP$
$HL = TL$

TEOREMA DE CARNOT

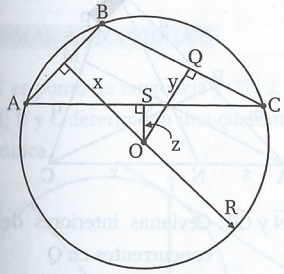
1er CASO: Triángulo acutángulo



Si: R y r son las medidas del circunradio e inradio.

$x + y + z = R + r$

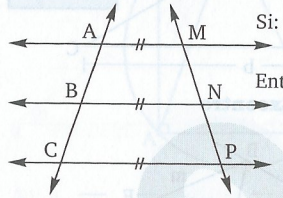
2do CASO: Triángulo obtusángulo



$x + y - z = R + r$

PROPORCIONALIDAD DE SEGMENTOS

TEOREMA DE TALES



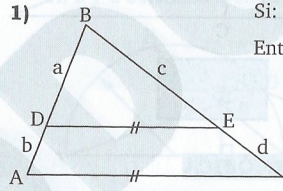
Si: $\overleftrightarrow{AM} // \overleftrightarrow{BN} // \overleftrightarrow{CP}$

Entonces:

$$\frac{AB}{BC} = \frac{MN}{NP}$$

Corolarios:

1)

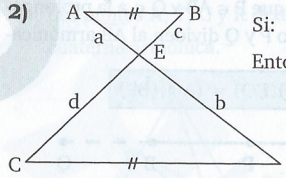


Si: $\overline{DE} // \overline{AC}$

Entonces:

$$\frac{a}{b} = \frac{c}{d}$$

2)



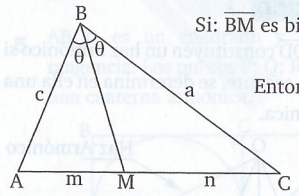
Si: $\overline{AB} // \overline{CD}$

Entonces:

$$\frac{a}{b} = \frac{c}{d}$$

PROPIEDADES RELATIVAS A LA BISECTRICES DE UN TRIÁNGULO

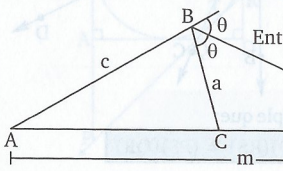
Teorema



Si: $\overline{BM}$ es bisectriz interior

Entonces: $\frac{c}{a} = \frac{m}{n}$

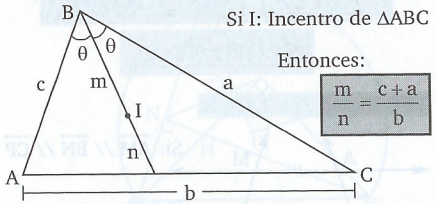
Teorema



Si: $\overline{BM}$ es bisectriz exterior

Entonces: $\frac{c}{a} = \frac{m}{n}$

Teorema del Incentro



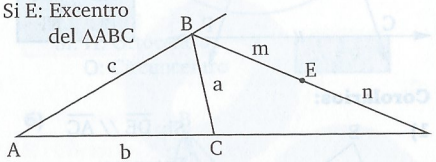
Si I: Incentro de ΔABC

Entonces:

$$\frac{m}{n}=\frac{c+a}{b}$$

Teorema del Excentro

Si E: Excentro del ΔABC



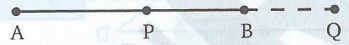
Entonces:

$$\frac{m}{n}=\frac{c-a}{b}$$

CUATERNA ARMÓNICA

Dado el $\overline{AB}$ tales que $P \in \overline{AB}$ y $Q \in$ a la prolongación de $\overline{AB}$, luego P y Q dividen al $\overline{AB}$ armónicamente si:

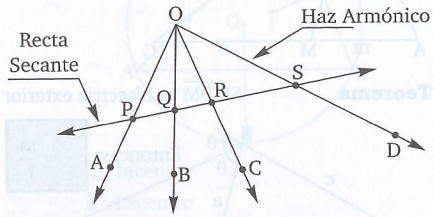
$$(AP)(BQ) = (AQ)(BP)$$



Donde: P y Q son los conjugados armónicos de A y B; además: A y B son los conjugados armónicos de P y Q.

HAZ ARMÓNICO

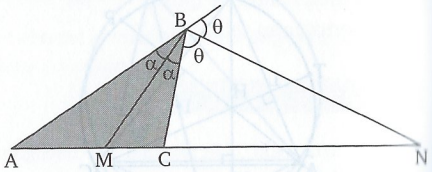
$\overrightarrow{OA}$; $\overrightarrow{OB}$; $\overrightarrow{OC}$ y $\overrightarrow{OD}$ constituyen un haz armónico si para toda recta secante, se determina en ella una cuaterna armónica.



Entonces se cumple que:

$$(PQ)(RS) = (PS)(QR)$$

TEOREMA

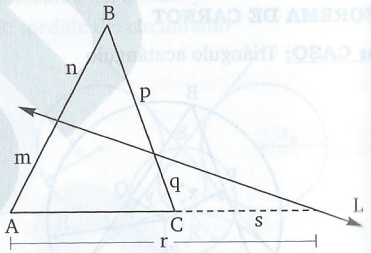


A – M – C – N: Forman una cuaterna armónica

Entonces:

$$\frac{AM}{MC}=\frac{AN}{CN} \quad \text{o} \quad (AM)(CN)=(MC)(AN)$$

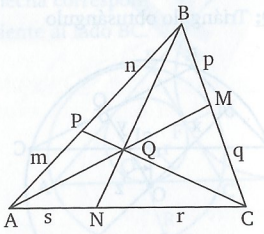
TEOREMA DE MENELAO



$\overleftarrow{L}$: Recta secante a $\overline{AB}$, $\overline{BC}$ y a la prolongación de $\overline{AC}$

$$mps = nqr$$

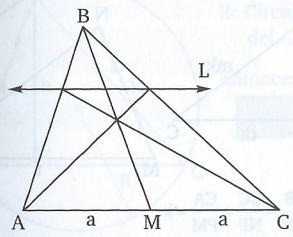
TEOREMA DE CEVA



$\overline{AM}$, $\overline{BN}$ y $\overline{CP}$: Cevianas interiores del ΔABC, concurrentes en Q

$$mpr = nqs$$

Corolario

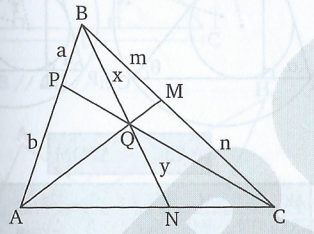


Si: $\overline{BM}$ es mediana

Entonces:

$$\overleftrightarrow{L} // \overline{AC}$$

TEOREMA DE VAN AUBEL



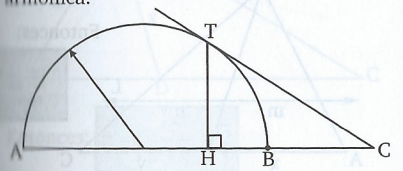
$\overline{AM}$, $\overline{BN}$ y $\overline{CP}$: Cevianas interiores del ΔABC concurrentes en Q

Entonces:

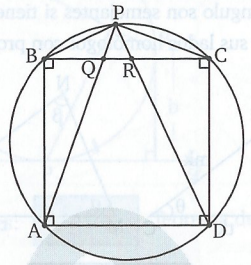
$$\frac{x}{y}=\frac{a}{b}+\frac{m}{n}$$

TEOREMAS ADICIONALES

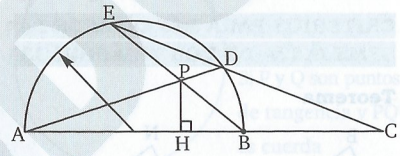
1. Si: T es punto de tangencia y $\overline{TH} \perp \overline{AB}$ A; H; B y C determinan una cuaterna armónica.



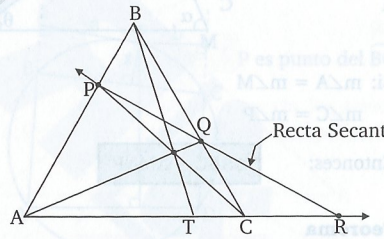
2. Si: ABCD es cuadrado y $P \in \widehat{BC}$; B; Q; R y C determinan una cuaterna armónica.



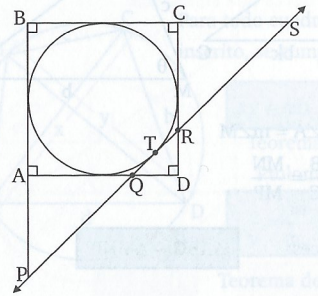
3. Si: $\overline{AD} \wedge \overline{EB} = \{P\}$ y $\overline{PH} \perp \overline{AB}$ entonces A; H; B y C determinan una cuaterna armónica.



4. Los puntos A; T; C y R determinan una cuaterna armónica.

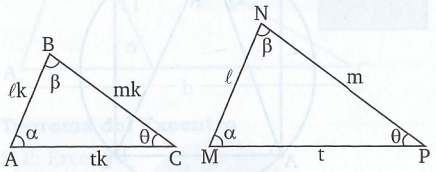


5. ABCD es un cuadrado y T es punto de tangencia. Los puntos P; Q; R y S determinan una cuaterna armónica.



SEMEJANZA DE TRIÁNGULOS

Dos triángulo son semejantes si tienen la misma forma y sus lados homólogos son proporcionales

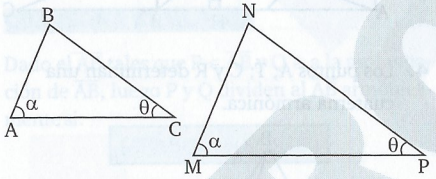


$\Delta ABC \sim \Delta MNP$

~: Se lee semejante

CRITERIOS PARA ESTABLECER LA SEMEJANZA DE DOS TRIÁNGULOS

Teorema

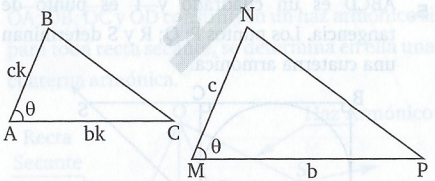


Si: $m\angle A = m\angle M$

$m\angle C = m\angle P$

Entonces: $\Delta ABC \sim \Delta MNP$

Teorema

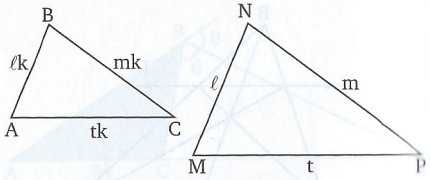


Si: $m\angle A = m\angle M$

$\frac{AB}{AC} = \frac{MN}{MP}$

Entonces: $\Delta ABC \sim \Delta MNP$

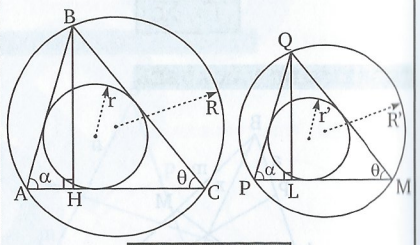
Teorema



Si: $\frac{AB}{MN} = \frac{BC}{NP} = \frac{CA}{PM} = k$

Entonces: $\Delta ABC \sim \Delta MNP$

NOTA

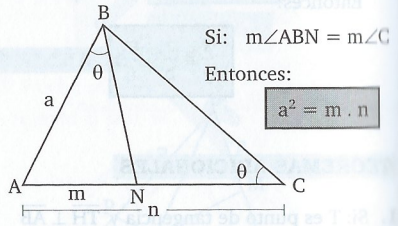


$\Delta ABC \sim \Delta PQM$

$$\frac{AB}{PQ} = \frac{BC}{QM} = \frac{AC}{PM} = \frac{BH}{QL} = \frac{r}{r'} = \frac{R}{R'} = \frac{2p_{ABC}}{2p_{PQM}} = \ldots = k$$

TEOREMAS

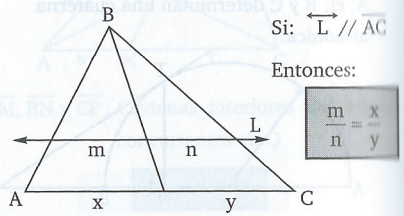
1)



Si: $m\angle ABN = m\angle C$

Entonces: $a^2 = m \cdot n$

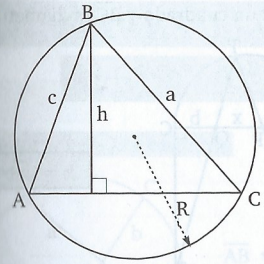
2)



Si: $\overleftrightarrow{L} \parallel \overline{AC}$

Entonces: $\frac{m}{n} = \frac{x}{y}$

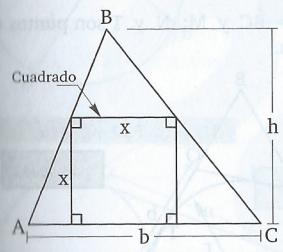
3)



R: Circunradio del ΔABC

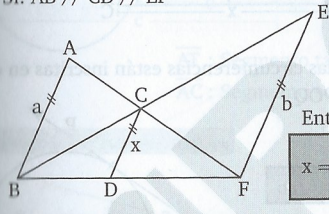
Entonces: $ac = 2Rh$

4) Lado del cuadrado inscrito en el triángulo



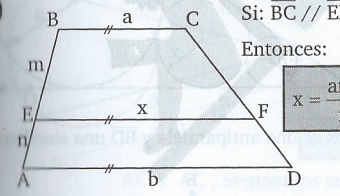
$x = \frac{bh}{b+h}$

5) Si: $\overline{AB} \parallel \overline{CD} \parallel \overline{EF}$



Entonces: $x = \frac{ab}{a+b}$

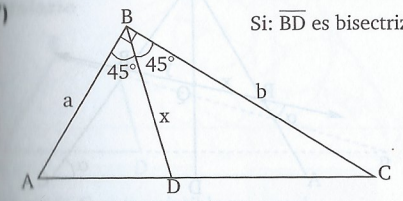
6)



Si: $\overline{BC} \parallel \overline{EF} \parallel \overline{AD}$

Entonces: $x = \frac{an+bm}{n+m}$

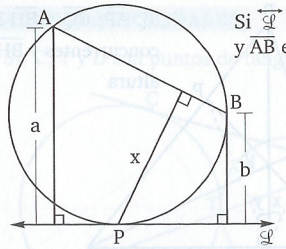
7)



Si: $\overline{BD}$ es bisectriz

Entonces: $x = \left(\frac{ab}{a+b}\right)\sqrt{2}$

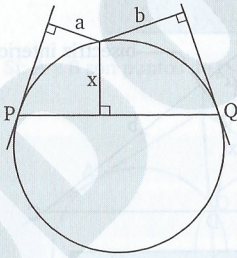
8)



Si $\overleftrightarrow{\mathcal{L}}$ es tangente y $\overline{AB}$ es cuerda

Entonces: $x = \sqrt{ab}$ Teorema de Pappus

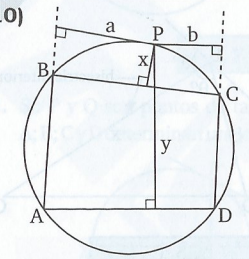
9)



Si P y Q son puntos de tangencia y $\overline{PQ}$ es cuerda

Entonces: $x^2 = ab$ Teorema de Pappus

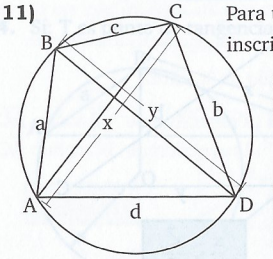
10)



P es punto del $\overset{\frown}{BC}$

Entonces: $ab = xy$ Teorema de Pappus

11)



Para todo cuadrilátero inscrito, se cumple:

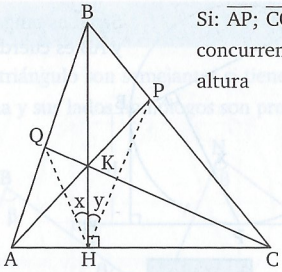
$xy = ab + cd$

Teorema de Ptolomeo

$\frac{x}{y} = \frac{ad+bc}{ac+bd}$

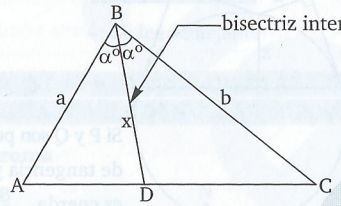
Teorema de Viette

12) Si: $\overline{AP}$; $\overline{CQ}$ y $\overline{BH}$ son concurrentes y $\overline{BH}$ es altura



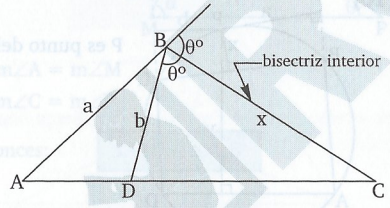
Entonces: $x = y$ Teorema de Blanchet

13)



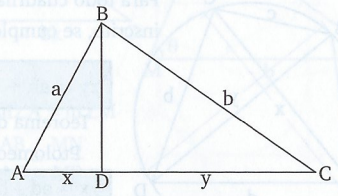
$$x = \left(\frac{2ab}{a+b}\right).\cos\alpha$$

14)



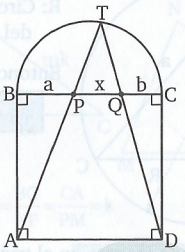
$$x = \left(\frac{2ab}{a-b}\right).\cos\alpha$$

15) Si: $\overline{BD}$ es una simediana



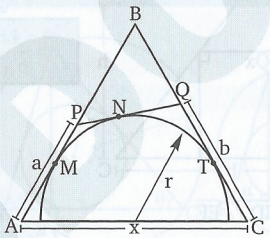
$$\frac{x}{y} = \frac{a^2}{b^2}$$

16) Si: ABCD es un cuadrado y $\overline{BC}$ es diámetro.



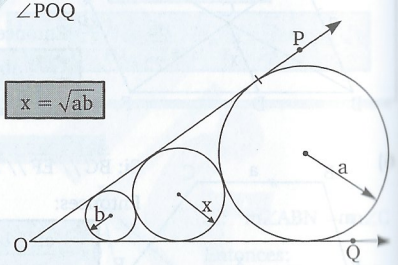
$$x = \sqrt{ab}$$

17) Si: AB = BC y M; N y T son puntos de tangencia.



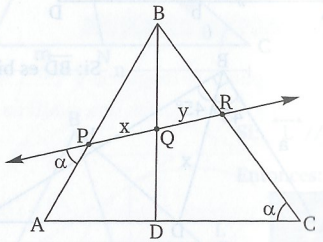
$$x = 2\sqrt{ab}$$

18) Si las circunferencias están inscritas en el $\angle POQ$



$$x = \sqrt{ab}$$

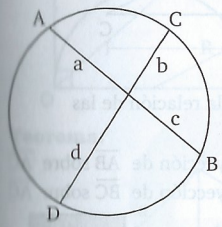
19) Si $\overline{PR}$ es una antiparalela y $\overline{BD}$ una sinediana.



$$x = y$$

RELACIONES MÉTRICAS EN LA CIRCUNFERENCIA

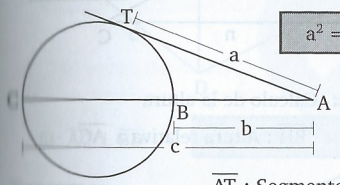
Teorema de las cuerdas



$\overline{AB}$ y $\overline{CD}$: cuerdas

$$ac = bd$$

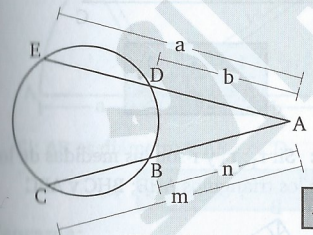
Teorema de la tangente



$$a^2 = bc$$

$\overline{AT}$: Segmento tangente
$\overline{AC}$: Segmento secante

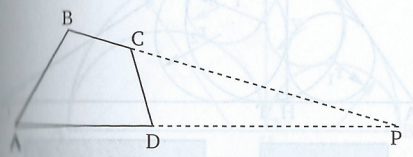
Teorema de las secantes



$$ab = mn$$

$\overline{AE}$ y $\overline{AC}$: Segmentos secantes

Corolario

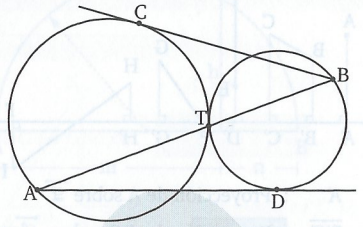


Si el $\square ABCD$ es inscriptible se cumple:

$$PB \cdot PC = PA \cdot PD$$

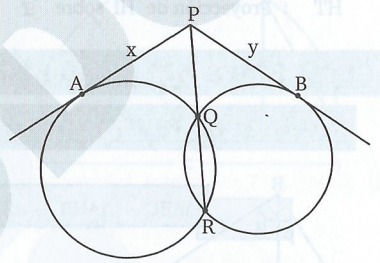
TEOREMA ADICIONALES

1. Si: C; T y D son puntos de tangencia.



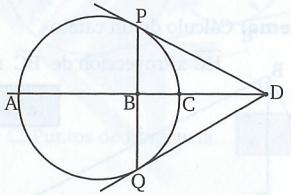
$$(AB)^2 = (BC)^2 + (AD)^2$$

2. Si: A y B son puntos de tangencia.

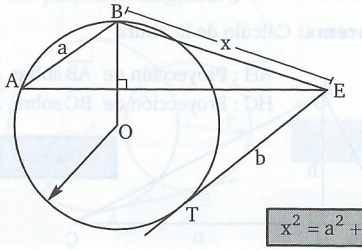


$$x = y$$

3. Si: P y Q son puntos de tangencia, los puntos A; B; C y D determinan una cuaterna armónica.

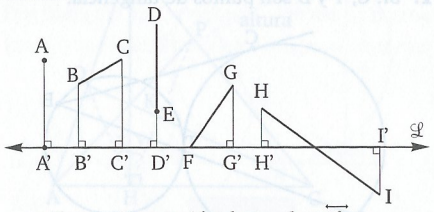


4. Si: T es punto de tangencia.



$$x^2 = a^2 + b^2$$

PROYECCIÓN ORTOGONAL SOBRE UNA RECTA



A' : Proyección de A sobre $\overleftrightarrow{\mathcal{L}}$

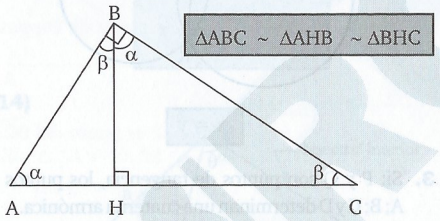
$\overline{B'C'}$: Proyección de $\overline{BC}$ sobre $\overleftrightarrow{\mathcal{L}}$

D' : Proyección de $\overline{DE}$ sobre $\overleftrightarrow{\mathcal{L}}$

$\overline{F'G'}$: Proyección de $\overline{FG}$ sobre $\overleftrightarrow{\mathcal{L}}$

$\overline{H'I'}$: Proyección de $\overline{HI}$ sobre $\overleftrightarrow{\mathcal{L}}$

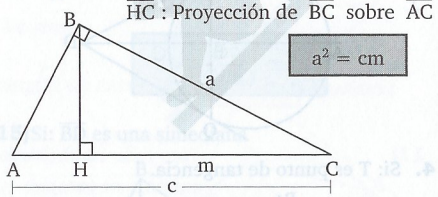
RELACIONES MÉTRICAS EN EL TRIÁNGULO RECTÁNGULO



$\Delta ABC \sim \Delta AHB \sim \Delta BHC$

Teorema: Cálculo de un cateto

$\overline{HC}$: Proyección de $\overline{BC}$ sobre $\overline{AC}$

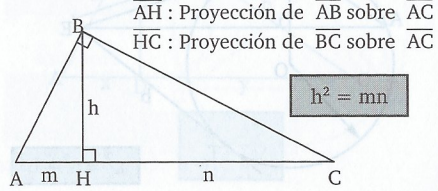


$a^2 = cm$

Teorema: Cálculo de la altura

$\overline{AH}$: Proyección de $\overline{AB}$ sobre $\overline{AC}$

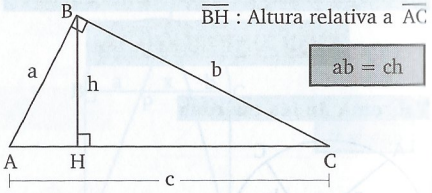
$\overline{HC}$: Proyección de $\overline{BC}$ sobre $\overline{AC}$



$h^2 = mn$

Teorema: Cálculo del producto de catetos

$\overline{BH}$: Altura relativa a $\overline{AC}$

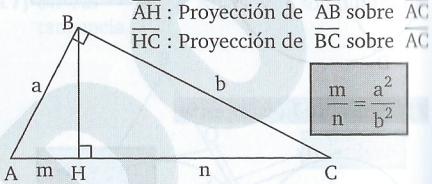


$ab = ch$

Teorema: Cálculo de la relación de las proyecciones

$\overline{AH}$: Proyección de $\overline{AB}$ sobre $\overline{AC}$

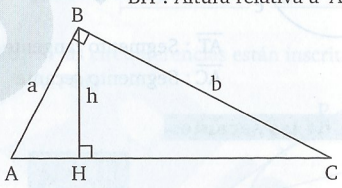
$\overline{HC}$: Proyección de $\overline{BC}$ sobre $\overline{AC}$



$\frac{m}{n} = \frac{a^2}{b^2}$

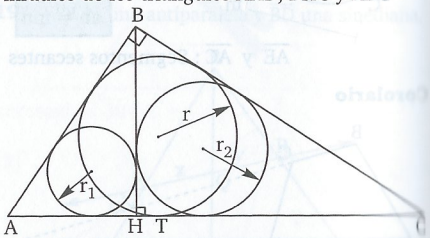
Teorema: Cálculo de la altura

$\overline{BH}$: Altura relativa a $\overline{AC}$



$\frac{1}{h^2} = \frac{1}{a^2} + \frac{1}{b^2}$

Teorema: Si: r_1; r_2 y r son las medidas de los inradios de los triángulos AHB; BHC y ABC

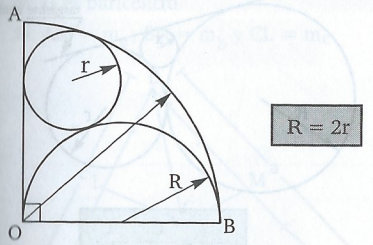


$HT = r_2 - r_1$

$r^2 = (r_1)^2 + (r_2)^2$

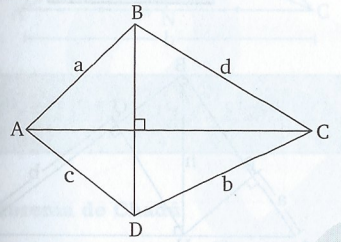
$BH = r_1 + r_2 + r$

Teorema:



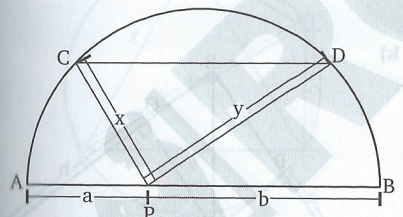
$R = 2r$

Teorema:



Si: $\overline{AC} \perp \overline{BD} \rightarrow$ $a^2 + b^2 = c^2 + d^2$

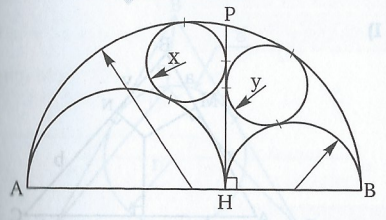
Teorema:



Si: AB es diámetro y AB // CD

$\rightarrow$ $a^2 + b^2 = c^2 + d^2$

Teorema:

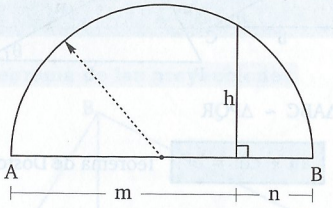


Si: AB es diámetro y PH ⊥ AB

$\rightarrow$ $x = y$

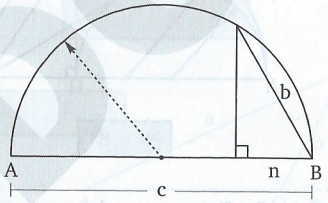
TEOREMAS ADICIONALES

a)



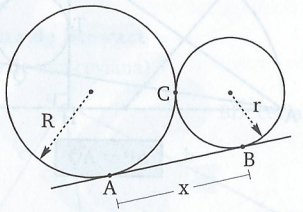
$\overline{AB}$: Diámetro $h^2 = mn$

b)



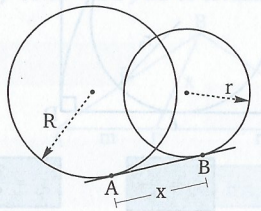
$\overline{AB}$: Diámetro $b^2 = cn$

c)



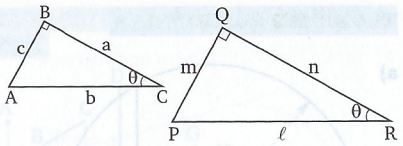
A, B y C: Puntos de tangencia $x = 2\sqrt{Rr}$

d) Si las circunferencias son ortogonales, A y B son puntos de tangencia



$x = \sqrt{2Rr}$

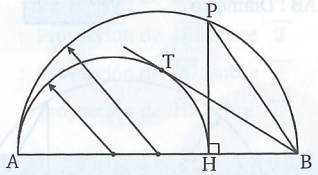
e)



Si: $\Delta ABC \sim \Delta PQR$

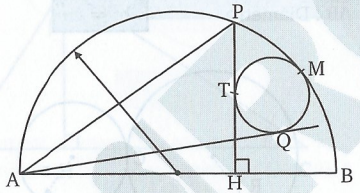
$an + cm = b\ell$ Teorema de Dostor

f) Si: T es punto de tangencia.



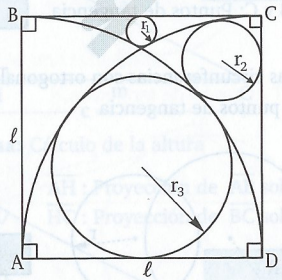
$PB = BT$

g) Si: M; Q y T son puntos de tangencia.



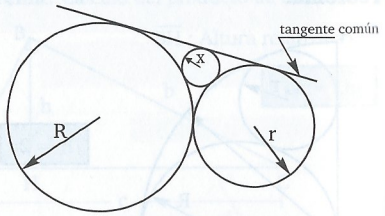
$AP = AQ$

h) ABOD en una cuadrado.



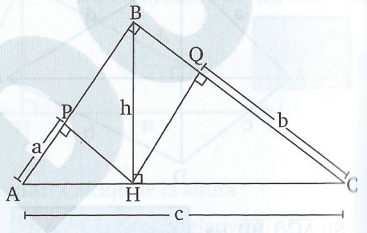
$r_1 = \frac{\ell}{16}$ $r_2 = \frac{\ell}{6}$ $r_3 = \frac{3\ell}{8}$

i)



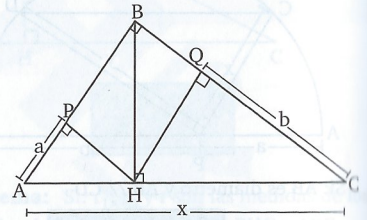
$$\frac{1}{\sqrt{x}} = \frac{1}{\sqrt{r}} + \frac{1}{\sqrt{R}}$$

j)



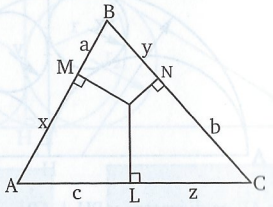
$h^3 = abc$

k)



$$\sqrt[3]{x^2} = \sqrt[3]{a^2} + \sqrt[3]{b^2}$$

l)

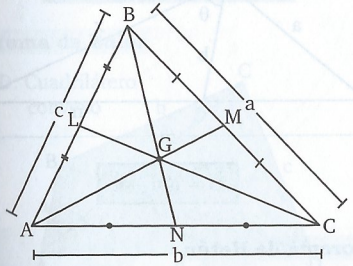


$x^2 + y^2 + z^2 = a^2 + b^2 + c^2$

m) Teorema de Booht

Si: G es baricentro

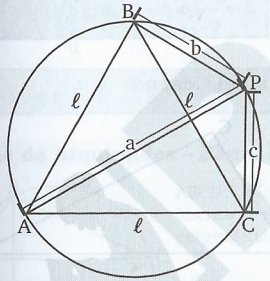
Si: $AM = m_a$; $BN = m_b$ y $CL = m_c$



$$\frac{(m_a)^2 + (m_b)^2 + (m_c)^2}{3} = \frac{a^2 + b^2 + c^2}{4}$$

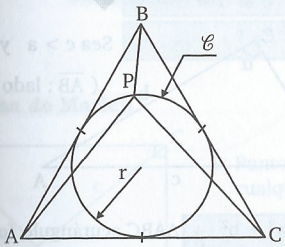
n) Teorema de Chadú

Si: $AB = BC = AC$ y $P \in \overset{\frown}{BC}$



$a = b + c$ $a^2 + b^2 + c^2 = 2\ell^2$

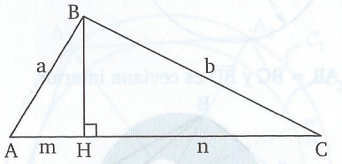
ñ) Si: $AB = BC = AC$ y $P \in \mathscr{C}$



$(PA)^2 + (PB)^2 + (PC)^2 = 15r^2$

RELACIONES MÉTRICAS EN TRIÁNGULOS OBLICUÁNGULOS

Teorema de las proyecciones

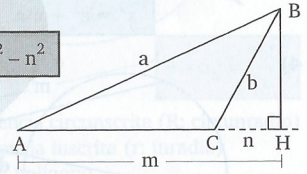


$\overline{AH}$: Proyección de $\overline{AB}$ sobre $\overline{AC}$

$\overline{HC}$: Proyección de $\overline{BC}$ sobre $\overline{AC}$

$b^2 - a^2 = n^2 - m^2$

$a^2 - b^2 = m^2 - n^2$

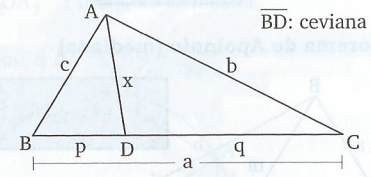


$\overline{AH}$: Proyección de $\overline{AB}$ sobre $\overline{AC}$

$\overline{HC}$: Proyección de $\overline{BC}$ sobre $\overline{AC}$

Teorema de Stewart

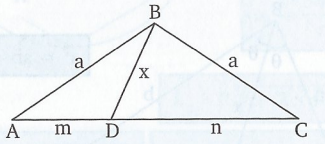
(cálculo de una ceviana)



$c^2q + b^2p = x^2a + pqa$

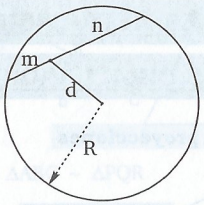
Corolarios:

1) Si $AB = BC$ y $\overline{BD}$ es ceviana interior



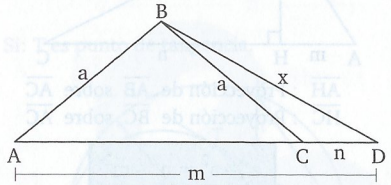
$x^2 = a^2 - mn$

2)



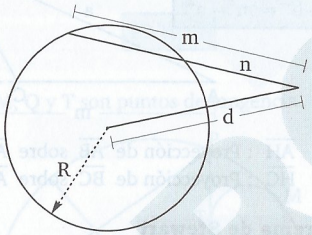
$$mn = R^2 - d^2$$

3) Si AB = BC y $\overline{BD}$ es ceviana interior



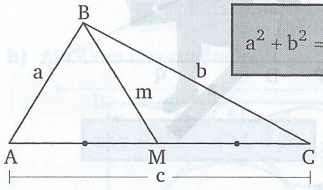
$$x^2 = a^2 + mn$$

4)



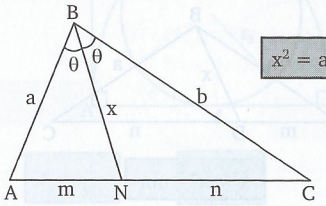
$$mn = d^2 - R^2$$

Teorema de Apolonio (mediana)



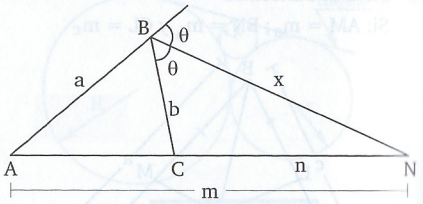
$$a^2 + b^2 = 2m^2 + \frac{c^2}{2}$$

Teorema de la bisectriz (interior)



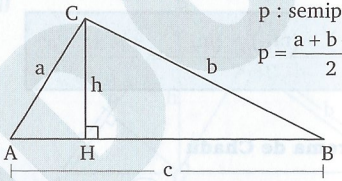
$$x^2 = ab - mn$$

Teorema de la bisectriz (exterior)



$$x^2 = mn - ab$$

Teorema de Herón

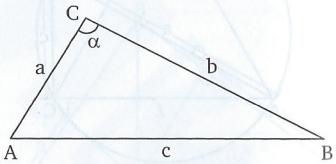


p : semiperímetro

$$p = \frac{a+b+c}{2}$$

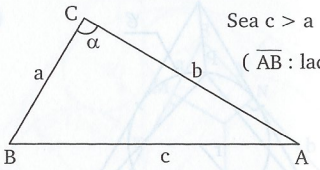
$$h = \frac{2}{c}\sqrt{p(p-a)(p-b)(p-c)}$$

Teorema del Coseno $\forall\ \alpha$



$$c^2 = a^2 + b^2 - 2ab\cos\alpha$$

Naturaleza de un triángulo



Sea c > a y c > b

($\overline{AB}$: lado mayor)

Si $c^2 < a^2 + b^2 \rightarrow$ ΔABC acutángulo ($\alpha < 90°$)

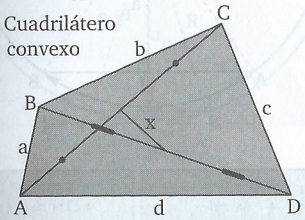
Si $c^2 = a^2 + b^2 \rightarrow$ ΔABC rectángulo ($\alpha = 90°$)

Si $c^2 > a^2 + b^2 \rightarrow$ ΔABC obtusángulo ($\alpha > 90°$)

RELACIONES MÉTRICAS EN CUADRILÁTEROS

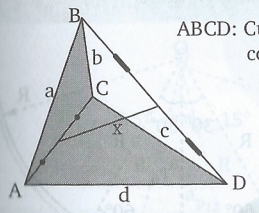
Teorema de Euler

ABCD: Cuadrilátero convexo



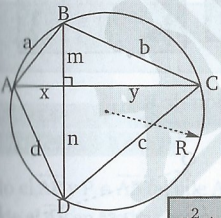
$$a^2 + b^2 + c^2 + d^2 = (AC)^2 + (BD)^2 + 4x^2$$

ABCD: Cuadrilátero no convexo



$$a^2 + b^2 + c^2 + d^2 = (AC)^2 + (BD)^2 + 4x^2$$

Teorema de Arquímides – Faure

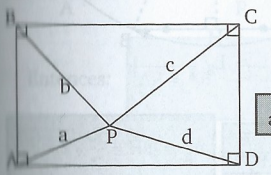


ABCD: Cuadrilátero inscrito de diagonales perpendiculares

$$a^2 + c^2 = b^2 + d^2 = 4R^2$$

$$x^2 + m^2 + y^2 + n^2 = 4R^2$$

Teorema de Marlen

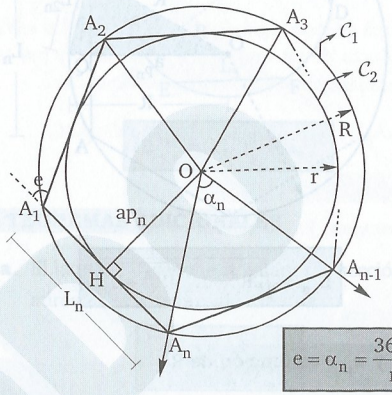


P: Punto cualquiera

$$a^2 + c^2 = b^2 + d^2$$

POLÍGONO REGULAR

Polígono que es equiángulo y equilátero a la vez. Todo polígono regular es inscriptible y circunscriptible a dos circunferencias concéntricas.



$$e = \alpha_n = \frac{360°}{n}$$

C_1 : Circunferencia circunscrita (R: circunradio)

C_2 : Circunferencia inscrita (r: inradio)

O : Centro del polígono

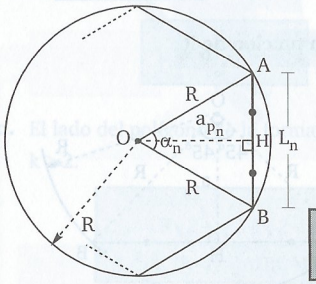
L_n : Longitud del lado del polígono regular de "n" lados

a_{p_n} : Longitud de la apotema del polígono regular de "n" lados

$\angle A_nOA_{n-1}$: Ángulo central del polígono regular

ΔA_2OA_3 : Triángulo elemental

Cálculo de "L_n" y "a_{p_n}"

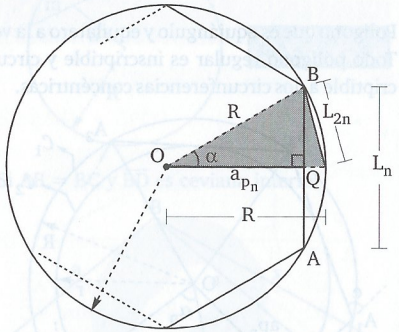


$$\alpha_n = \frac{360°}{n}$$

$$L_n = R\sqrt{2 - 2\cos\alpha_n}$$

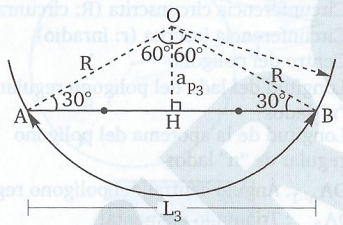
$$a_{p_n} = \frac{1}{2}\sqrt{4R^2 - L_n^2} \qquad a_{p_n} = \frac{R}{2}\sqrt{2 + 2\cos\alpha_n}$$

Cálculo de "L_{2n}"



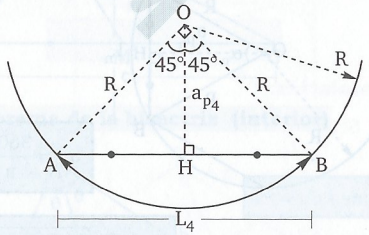
$$L_{2n} = \sqrt{2R^2 - R\sqrt{4R^2 - L_n^2}}$$

L_3 y a_{p_3} en función de R



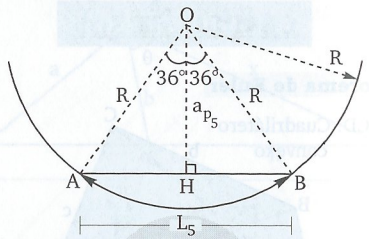
$$AB = L_3 = R\sqrt{3} \qquad a_{p_3} = OH = \frac{R}{2}$$

L_4 y a_{p_4} en función de R



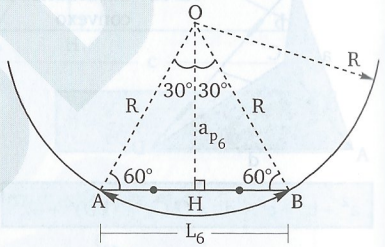
$$L_4 = AB = R\sqrt{2} \qquad a_{p_4} = OH = \frac{R\sqrt{2}}{2}$$

L_5 y a_{p_5} en función de R



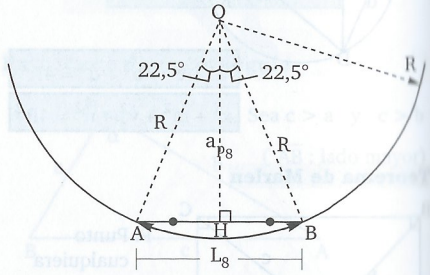
$$L_5 = AB = \frac{R}{2}\sqrt{10 - 2\sqrt{5}} \qquad a_{p_5} = OH = \frac{R}{4}(\sqrt{5} + 1)$$

L_6 y a_{p_6} en función de R



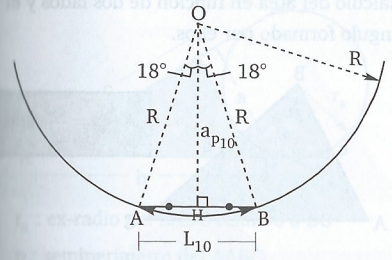
$$L_6 = AB = R \qquad a_{p_6} = OH = \frac{R}{2}\sqrt{3}$$

L_8 y a_{p_8} en función de R



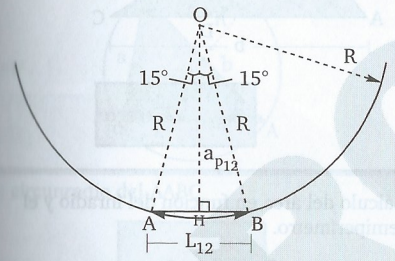
$$L_8 = AB = R\sqrt{2 - \sqrt{2}} \qquad a_{p_8} = OH = \frac{R}{2}\sqrt{2 + \sqrt{2}}$$

L_{10} y $a_{p_{10}}$ en función de R



$$L_{10} = AB = \frac{R}{2}(\sqrt{5} - 1) \qquad a_{p_{10}} = OH = \frac{R}{4}\sqrt{10 + 2\sqrt{5}}$$

L_{12} y $a_{p_{12}}$ en función de R

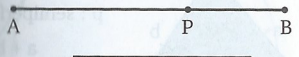


$$L_{12} = AB = R\sqrt{2 - \sqrt{3}} \qquad a_{p_{12}} = OH = \frac{R}{2}\sqrt{2 + \sqrt{3}}$$

DIVISIÓN DE UN SEGMENTO EN MEDIA Y EXTREMA RAZÓN

Dado el $\overline{AB}$ y $P \in \overline{AB}$ tal que $AB > PB$, entonces P divide al $\overline{AB}$ en media y extrema razón si:

$$(AP)^2 = (AB)(PB)$$

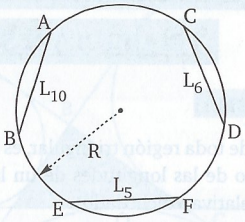


Entonces: $AP = AB\left(\frac{\sqrt{5} - 1}{2}\right)$

$\overline{AP}$: Sección áurea de $\overline{AB}$

$\overline{PB}$: Sección áurea de $\overline{AP}$

TEOREMA



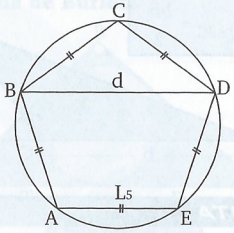
$$(L_5)^2 = (L_{10})^2 + (L_6)^2$$

TEOREMAS ADICIONALES

a. El lado del decágono regular es la sección áurea de su circunradio.

$$L_{10} = R\left(\frac{\sqrt{5} - 1}{2}\right)$$

b. El lado del pentágono regular es la sección áurea de una de sus diagonales.



$$L_5 = d\left(\frac{\sqrt{5} - 1}{2}\right)$$

c. El lado del polígono de la forma L_{2^k} donde $k \geq 2$.

$$L_{2^k} = \underbrace{R\sqrt{2 - \sqrt{2 + \sqrt{2 + \sqrt{2}...}}}}_{(k-1)\text{ RADICALES}}$$

Ejemplo:

$L_8 = R\sqrt{2 - \sqrt{2}}$; $L_{16} = R\sqrt{2 - \sqrt{2 + \sqrt{2}}}$;

$L_{32} = R\sqrt{2 - \sqrt{2 + \sqrt{2 + \sqrt{2}}}}$

ÁREAS DE REGIONES TRIANGULARES

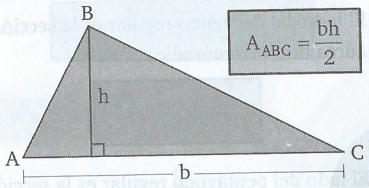
TEOREMA

El área de toda región triangular, es igual al semi producto de las longitudes de un lado y por la altura relativa a dicho lado.

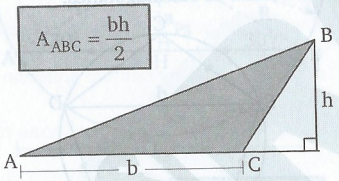
Sea A_{ABC}: Área de la región triangular ABC.

TEOREMA FUNDAMENTAL

Teorema: Para toda región triangular



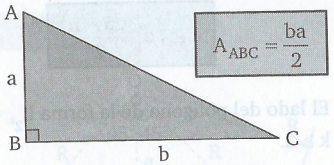
$$A_{ABC} = \frac{bh}{2}$$



$$A_{ABC} = \frac{bh}{2}$$

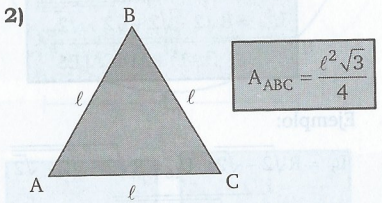
NOTA

1)



$$A_{ABC} = \frac{ba}{2}$$

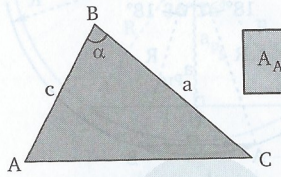
2)



$$A_{ABC} = \frac{\ell^2\sqrt{3}}{4}$$

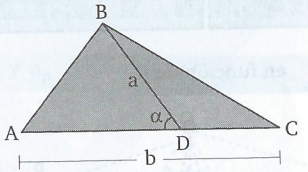
TEOREMA TRIGONOMÉTRICO

Cálculo del área en función de dos lados y el ángulo formado por ellos.



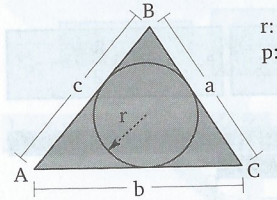
$$A_{ABC} = \frac{ac}{2}\operatorname{sen}\alpha$$

NOTA



$$A_{ABC} = \frac{ab}{2}\operatorname{sen}\alpha$$

Cálculo del área en función del inradio y el semiperímetro.

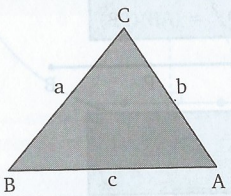


r: inradio de ΔABC
p: semiperímetro del ΔABC

$$A_{ABC} = pr$$

TEOREMA DE HERÓN

Cálculo del área en función de los lados

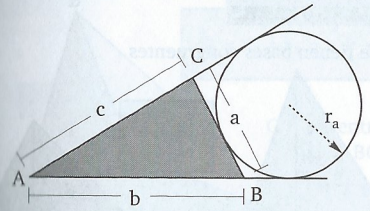


p : semiperímetro

$$p = \frac{a+b+c}{2}$$

$$A_{ABC} = \sqrt{p(p-a)(p-b)(p-c)}$$

Cálculo del área en función del exradio

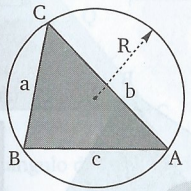


r_a : ex-radio del ΔABC relativo a $\overline{BC}$

p : semiperímetro del ΔABC

$$A_{ABC} = r_a(p-a)$$

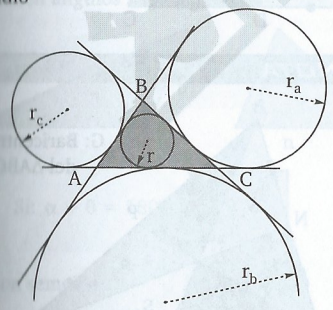
Cálculo del área en función del circunradio



R: circunradio del ΔABC

$$A_{ABC} = \frac{abc}{4R}$$

Cálculo del área en función de los exradios y el inradio

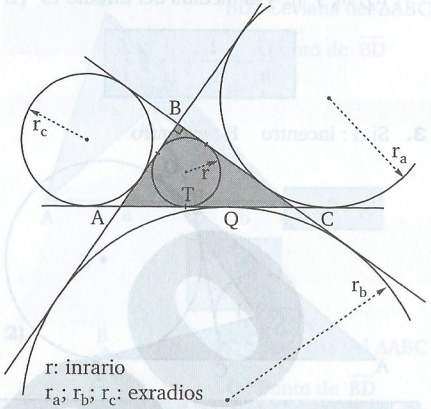


r: inrario
r_a; r_b; r_c: exradios

$$\frac{1}{r_a}+\frac{1}{r_b}+\frac{1}{r_c}=\frac{1}{r}$$

$$A_{ABC} = \sqrt{r_a \cdot r_b \cdot r_c \cdot r}$$

Teoremas para el triángulo rectángulo



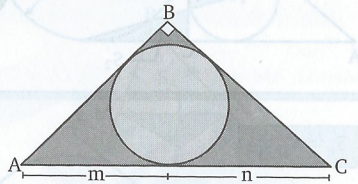
r: inrario
r_a; r_b; r_c: exradios

$$A_{ABC} = (AT)(TC) \qquad A_{ABC} = r \cdot r_b$$

$$A_{ABC} = (AQ)(QC) \qquad A_{ABC} = r_a \cdot r_c$$

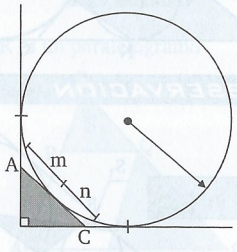
TEOREMAS ADICIONALES

1. Teorema de Burlet.



$$S(ABC) = mn$$

OBSERVACIÓN

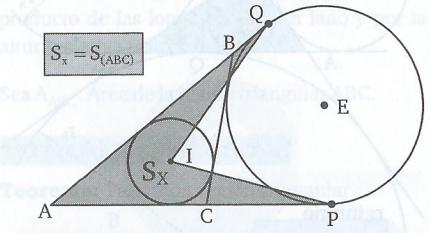


$$S(ABC) = mn$$

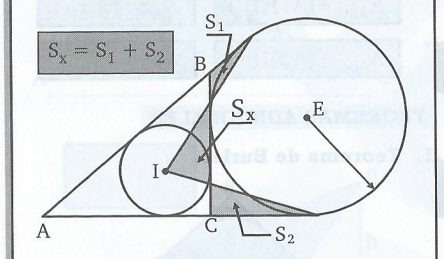
2. Si las alturas de un triángulo miden h_a; h_b y h_c y la medida del inradio es "r".

$$\frac{1}{h_a}+\frac{1}{h_b}+\frac{1}{h_c}=\frac{1}{r}$$

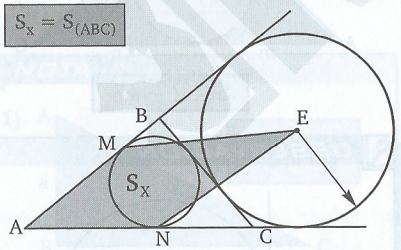
3. Si: I : incentro E : excentro



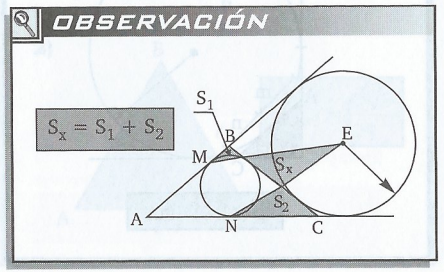
OBSERVACIÓN



4. $S_x = S_{(ABC)}$

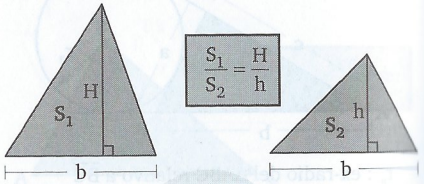


OBSERVACIÓN



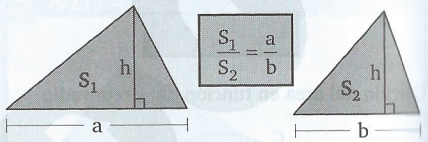
RELACIÓN ENTRE ÁREAS DE REGIONES TRIANGULARES

Si se tienen bases congruentes



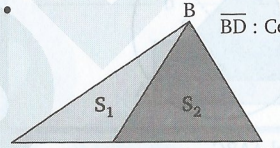
$$\frac{S_1}{S_2}=\frac{H}{h}$$

Si tienen alturas congruentes

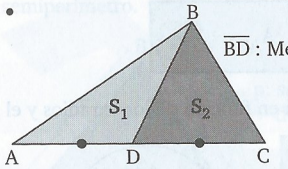


$$\frac{S_1}{S_2}=\frac{a}{b}$$

Corolario

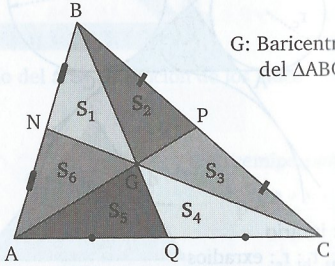
- 

$$\frac{S_1}{S_1}=\frac{a}{b}$$

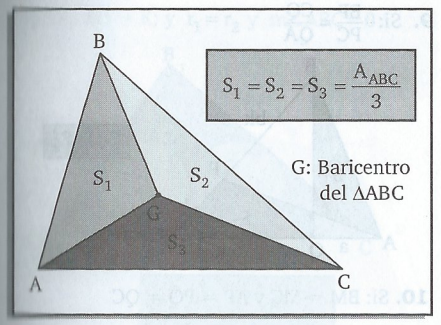
- 

$$S_1 = S_2$$

NOTA



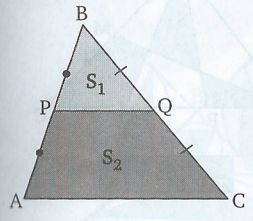
$$S_1 = S_2 = S_3 = S_4 = S_5 = S_6 = \frac{A_{ABC}}{6}$$



$$S_1 = S_2 = S_3 = \frac{A_{ABC}}{3}$$

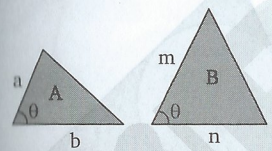
G: Baricentro del ΔABC

Si $\overline{PQ}$ es base media



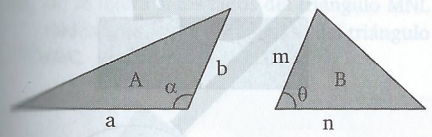
$$\frac{S_1}{S_2}=\frac{1}{3}$$

Si tienen un ángulo de igual medida



$$\frac{A}{B}=\frac{ab}{mn}$$

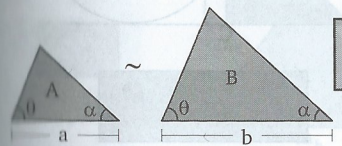
Si tienen ángulos suplementarios



Si: $\alpha + \theta = 180°$

$$\frac{A}{B}=\frac{ab}{mn}$$

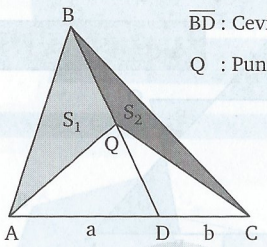
Si son semejantes



$$\frac{A}{B}=\frac{a^2}{b^2}$$

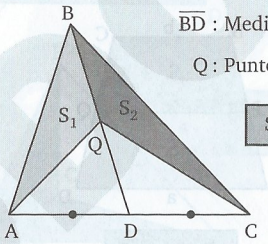
Teoremas Adicionales

1)



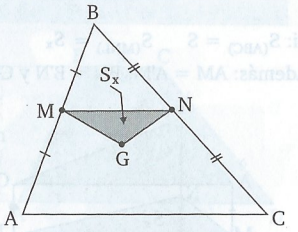
$$\frac{S_1}{S_2}=\frac{a}{b}$$

2)



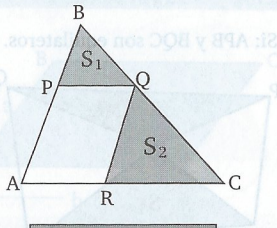
$$S_1 = S_2$$

3. G: Baricentro



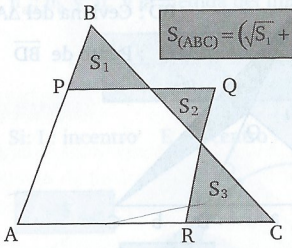
$$S_x = \frac{S_{(ABC)}}{12}$$

4. Si: APQR es un paralelogramo



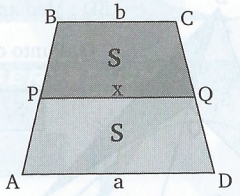
$$S_{(ABC)} = \left(\sqrt{S_1} + \sqrt{S_2}\right)^2$$

5. Si: APQR es un paralelogramo



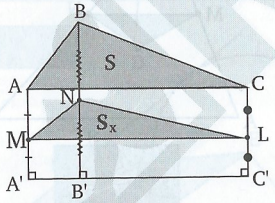
$$S_{(ABC)} = \left(\sqrt{S_1} + \sqrt{S_2} + \sqrt{S_3}\right)^2$$

6. Si las regiones trapeciales son equivalentes.
$S_{(ABCQ)} = S_{(APQD)}$



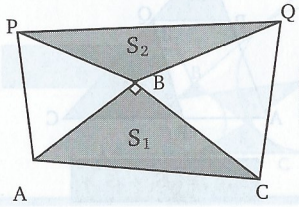
$$x = \sqrt{\frac{a^2 + b^2}{2}}$$

7. Si: $S_{(ABC)} = S$ $S_{(MNL)} = S_x$
Además: AM = A'M; BN = B'N y CL = C'L



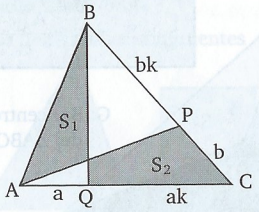
$$S_x = \frac{S}{2}$$

8. Si: APB y BQC son equilateros.



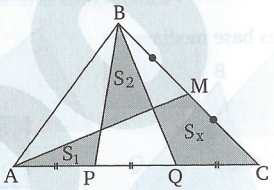
$$\frac{S_1}{S_2} = 2$$

9. Si: $\frac{BP}{PC} = \frac{CQ}{QA}$



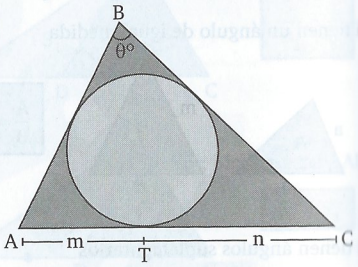
$$S_1 = S_2$$

10. Si: BM = MC y AP = PQ = QC



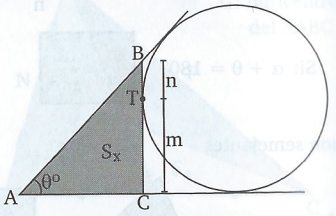
$$S_x = S_1 + S_2$$

11.



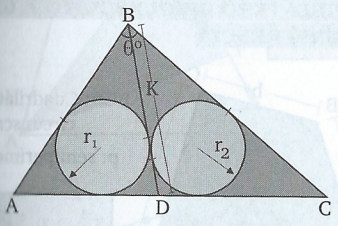
$$S_{(ABC)} = m.n\cot\left(\frac{\theta}{2}\right)$$

12.



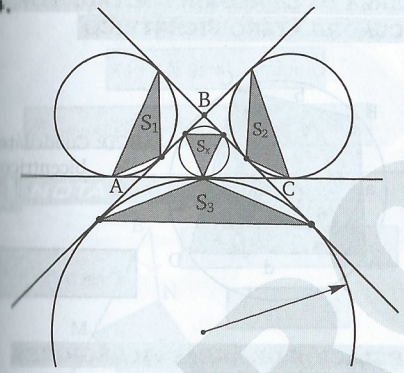
$$S_{(ABC)} = m.n\cot\left(\frac{\theta}{2}\right)$$

13. Si: BD = K; y $r_1 = r_2$ y $m\angle ABC = \theta$



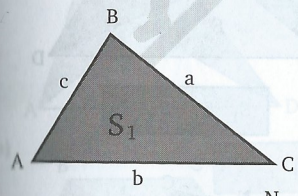
$$S_{(ABC)} = K^2.\cot\left(\frac{\theta}{2}\right)$$

14.

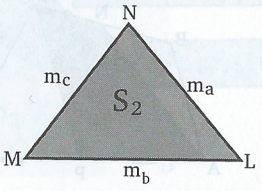


$$\frac{1}{S_x} = \frac{1}{S_1} + \frac{1}{S_2} + \frac{1}{S_3}$$

15. De la figura si los lados del triángulo MNL miden igual que las medianas del triángulo ABC.

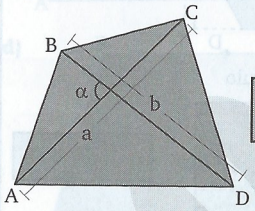


$$\frac{S_1}{S_2} = \frac{4}{3}$$

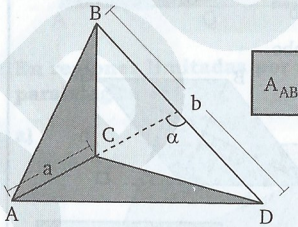


ÁREAS DE REGIONES CUADRANGULARES

FÓRMULA GENERAL (TRIGONOMÉTRICA)

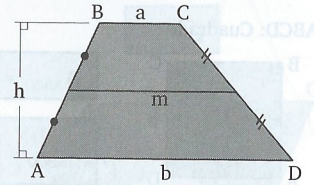


$$A_{ABCD} = \frac{ab}{2}\operatorname{sen}\alpha$$



$$A_{ABCD} = \frac{ab}{2}\operatorname{sen}\alpha$$

ÁREA DE UNA REGIÓN TRAPECIAL

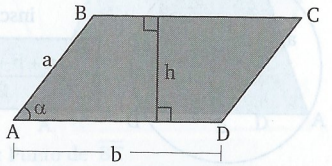


$$A_{ABCD} = \left(\frac{a+b}{2}\right)h$$

$$A_{ABCD} = mh$$

ÁREA DE UNA REGIÓN PARALELOGRÁMICA

Teorema

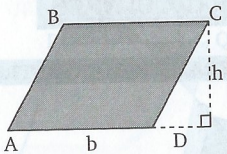


$$A_{ABCD} = bh$$

$$A_{ABCD} = ab\operatorname{sen}\alpha$$

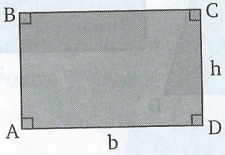
NOTA

ABCD: Romboide



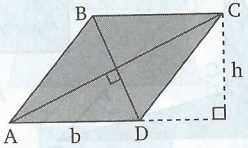
$A_{ABCD} = b \cdot h$

ABCD: Rectángulo



$A_{ABCD} = b \cdot h$

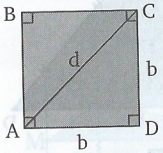
ABCD: Rombo



$A_{ABCD} = b \cdot h$

$A_{ABCD} = \frac{(AC)(BD)}{2}$

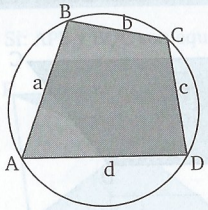
ABCD: Cuadrado



$A_{ABCD} = b^2$

$A_{ABCD} = \frac{d^2}{2}$

ÁREA DE LA REGIÓN LIMITADA POR UN CUADRILÁTERO INSCRITO O INSCRIPTIBLE

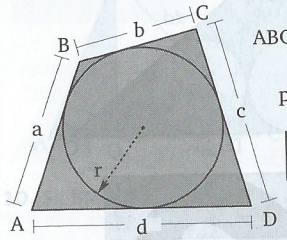


ABCD: Cuadrilátero inscrito o inscriptible

$p = \frac{a+b+c+d}{2}$

$A_{ABCD} = \sqrt{(p-a)(p-b)(p-c)(p-d)}$

ÁREA DE LA REGIÓN LIMITADA POR UN CUADRILÁTERO CIRCUNSCRITO A UNA CIRCUNFERENCIA

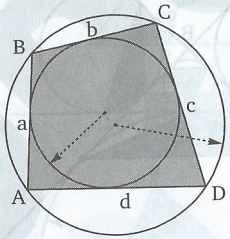


ABCD: Cuadrilátero circunscrito

p: semiperímetro

$A_{ABCD} = pr$

ÁREA DE LA REGIÓN LIMITADA POR UN CUADRILÁTERO BICÉNTRICO



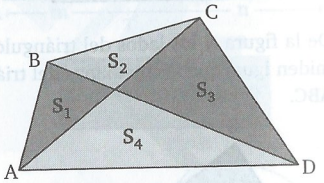
ABCD: Cuadrilátero bicentrico

$A_{ABCD} = \sqrt{abcd}$

RELACIÓN DE ÁREAS EN REGIONES CUADRANGULARES

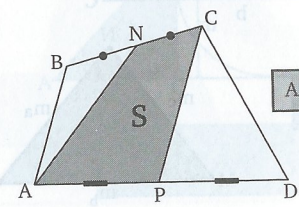
En regiones limitadas por un trapezoide

a)



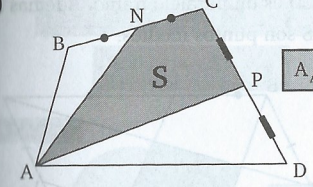
$S_1 \cdot S_3 = S_2 \cdot S_4$

b)



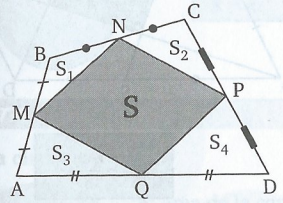
$A_{ABCD} = 2S$

c)



$A_{ABCD} = 2S$

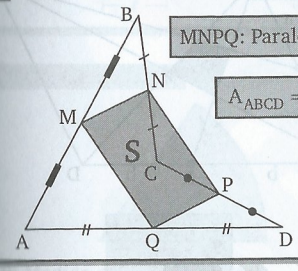
d)



MNPQ: Paralelogramo

$A_{ABCD} = 2S = 4(S_1 + S_4) = 4(S_2 + S_3)$

NOTA

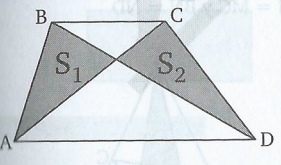


MNPQ: Paralelogramo

$A_{ABCD} = 2S$

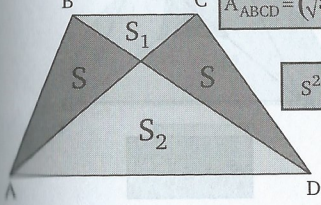
En regiones limitadas por un trapecio

a)



$S_1 = S_2$

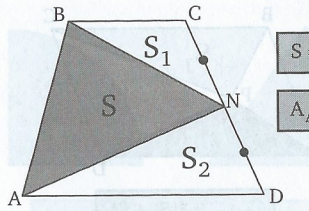
b)



$A_{ABCD} = \left(\sqrt{S_1} + \sqrt{S_2}\right)^2$

$S^2 = S_1 \cdot S_2$

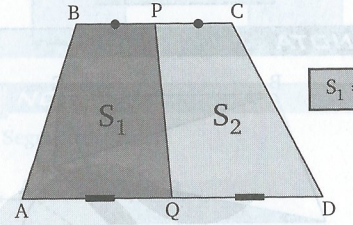
c)



$S = S_1 + S_2$

$A_{ABCD} = 2S$

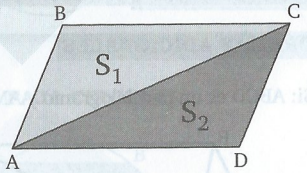
d)



$S_1 = S_2$

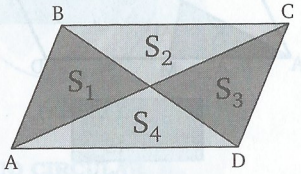
En regiones limitadas por un paralelogramo

a)



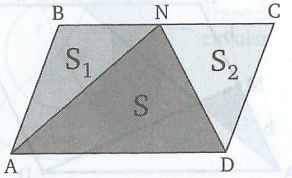
$S_1 = S_2$

b)



$S_1 = S_2 = S_3 = S_4$

c)

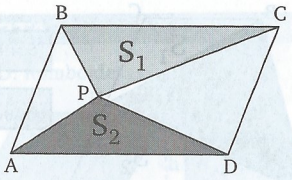


N: Punto de $\overline{BC}$

$A_{ABCD} = 2S$

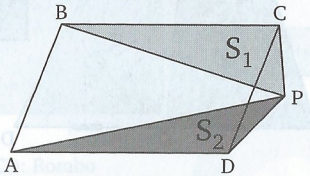
$S = S_1 + S_2$

d)



$$A_{ABCD} = 2(S_1 + S_2)$$

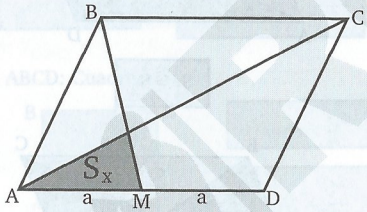
NOTA



$$A_{ABCD} = 2(S_1 + S_2)$$

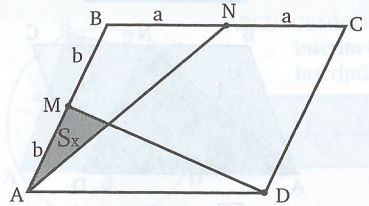
TEOREMAS ADICIONALES

1. Si: ABCD es un paralelogramo y AM = MD.



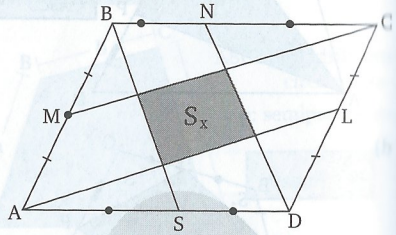
$$S_x = \frac{S_{(ABCD)}}{12}$$

2. Si: ABCD es un paralelogramo y AM = MB; BN = NC.



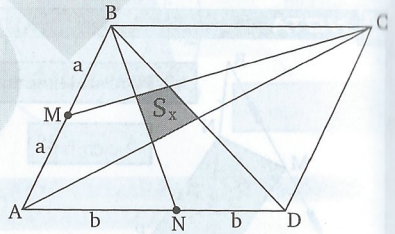
$$S_x = \frac{S_{(ABCD)}}{20}$$

3. Si: ABCD es un paralelogramo. Ademas M; N; L y S son puntos medios.



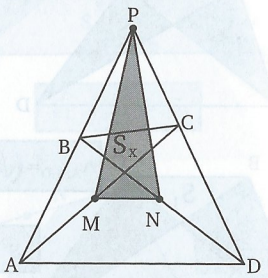
$$S_x = \frac{S_{(ABCD)}}{5}$$

4. Si: ABCD es un paralelogramo, además si AM = MB y AN = ND.



$$S_x = \frac{S_{(ABCD)}}{20}$$

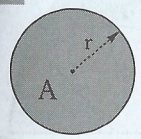
5. Si: AM = MC y BN = ND



$$S_x = \frac{S_{(ABCD)}}{4}$$

ÁREA DEL CÍRCULO Y SUS PARTES

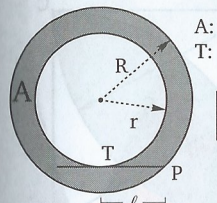
CÍRCULO



$$A = \pi r^2$$

NOTA

Corona circular

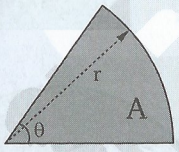


A: área de la corona
T: punto de contacto

$$A = \pi(R^2 - r^2)$$

$$A = \pi \ell^2$$

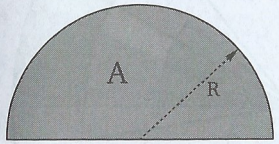
SECTOR CIRCULAR



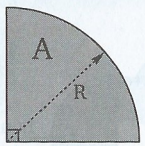
θ: Medida del ángulo central
A: área del sector circular

$$A = \left(\frac{\theta}{360°}\right)\pi r^2$$

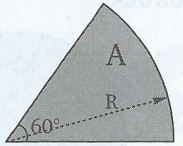
CASOS PARTICULARES



$$A = \frac{\pi R^2}{2}$$



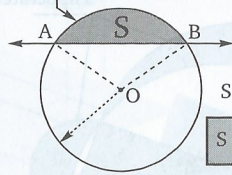
$$A = \frac{\pi R^2}{4}$$



$$A = \frac{\pi R^2}{6}$$

NOTA

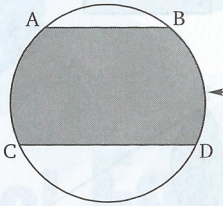
Segmento circular



S.C: Sector Circular

$$S = S_{S.C\,(AOB)} - S_{\Delta AOB}$$

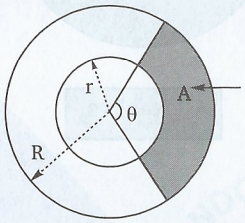
ZONA O FAJA CIRCULAR



Zona o faja circular

$(\overline{AB}\ //\ \overline{CD})$

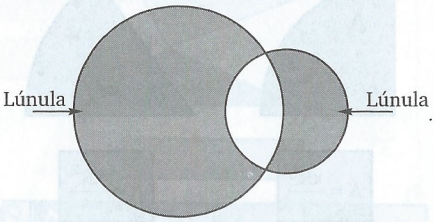
TRAPECIO CIRCULAR



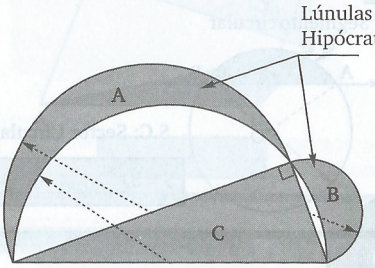
Trapecio circular

$$A = \left(\frac{\theta}{360°}\right)\pi(R^2 - r^2)$$

LÚNULA

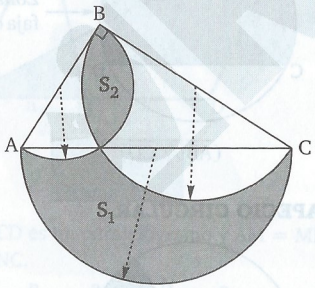


TEOREMA



$$C = A + B$$

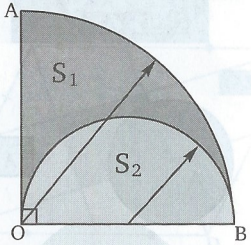
TEOREMA



$$A_{ABC} = S_1 - S_2$$

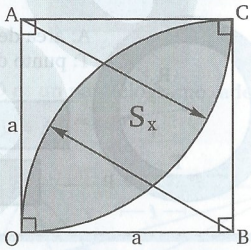
TEOREMAS ADICIONALES

1.



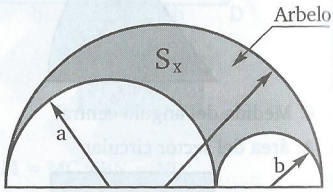
$$S_1 = S_2$$

2.



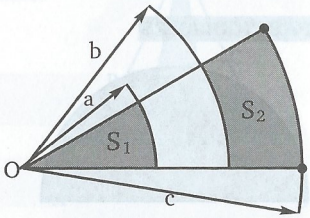
$$S_x = \frac{a^2}{2}(\pi - 2)$$

3.



$$S_x = \pi ab$$

4.



Si: $S_1 = S_2$ $\quad c^2 = a^2 + b^2$